TECHNISCHE UNIVERSITÄT DRESDEN

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Faculty of Electronic and Information Institute of Communication Engineering
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Master Thesis

Randomized Identification Codes with a Focus on K-Identification

Chongyang Zheng
Born on: 10th October 1997 in Xi’an
Course: Nanoelectronic Systems
Matriculation number: 5198083
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Supervisors
Juan A. Cabrera
Caspar von Lengerke
Supervising professor
Prof. Dr. h.c. Fitzek

TECHNISCHE UNIVERSITÄT DRESDEN

Fakultät Elektrotechnik und Informationstechnik
Institut für Nachrichtentechnik
Deutsche Telekom Professur Kommunikationsnetze
Prof. Dr.-Ing. Dr. h. c. Frank Fitzek

Task for Master Thesis of

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Topic: Randomized Identification Codes with a Focus on K-Identification

Task description:

Identification via channels (ID) asks receivers to decide whether a specific message was sent, rather than to reconstruct it. Randomized ID codes achieve doubly-exponential message sets in blocklength and enable vanishing Type-I (false-accept) and Type-II (miss) error probabilities at positive ID rate. K-Identification (K-ID) generalizes this: the receiver asks whether the transmitted message belongs to a set of K candidates. K-ID is relevant for goal/semantics-driven communication (e.g., digital twins), where the action depends only on whether the current state falls into a small set of actionable hypotheses, not on exact message recovery. This thesis will study randomized ID codes with emphasis on K-ID, and extend the ideas and evaluation from “Stopping the Data Flood: Post-Shannon traffic reduction in digital-twins applications” to the K-ID setting. The work should balance theory, algorithms, and software-driven experiments using our provided library.

The student is expected to:

Survey the state of the art:

- Core ID concepts: randomized encoders/acceptance regions, ID capacity, error exponents, and operational interpretations; contrast with Shannon transmission.
- Define K-ID precisely (membership in a K-set) and position it vs. related notions (multiple-object ID, generalized ID; deterministic vs randomized variants.

Formulate the problem & metrics for K-ID:

- Formal model (channel class, randomization/common randomness budget, decoder tests).
- Report both Type-I/Type-II errors and computational metrics (encode/decode time, seed length, memory), plus finite-blocklength behavior.

Replicate & extend the digital-twins baseline to K-ID:

- Reproduce key experiments/results from the “Stopping the Data Flood” study (baseline ID) with our software stack.
- Design K-ID experiments that reflect digital-twin quantifying traffic reduction vs. reliability/latency.

Use our software library to build a modular K-ID test bench:

- Implement randomized ID/K-ID encoders and decoders (acceptance region tests) as pluggable components.

Design and run a simulation campaign:

- Measure KPIs such as: Type-I/II error curves, empirical exponents, queries per symbol, and compute/memory usage. Quantify traffic reduction vs. reliability/latency

Deliver a comparative assessment with clear takeaways:

- When do systems benefit from K-ID?
- Tradeoffs among randomization, complexity, and reliability.

Supervisors:	Juan A. Cabrera, Caspar von Lengerke
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Prof. Dr.-Ing. Dr. h.c. Fitzek
Responsible Tutor

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Statement of authorship

I hereby certify that I have authored this document entitled *Randomized Identification Codes with a Focus on K-Identification* independently and without undue assistance from third parties. No other than the resources and references indicated in this document have been used. I have marked both literal and accordingly adopted quotations as such. During the preparation of this document I was only supported by the following persons:

Juan A. Cabrera
Caspar von Lengerke

Additional persons were not involved in the intellectual preparation of the present document. I am aware that violations of this declaration may lead to subsequent withdrawal of the academic degree.

Dresden, 7th February 2026

Chongyang Zheng

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Abstract

In networked control systems, a Digital Twin maintains a global state estimate to synchronize with a physical robot. Traditional synchronization relies on periodic full-state transmissions, which are often redundant because robots possess local sensors for state estimation. This thesis investigates K-Identification, a verification-based protocol where the Digital Twin transmits a compact verifier instead of the full payload. The robot uses this verifier to validate its local sensor data against the Digital Twin's global prediction within a defined semantic tolerance radius R.

The evaluation is conducted in three stages. First, an end-to-end latency framework compares truncated SHA-256 with Reed-Solomon-based identification. The results show that optimized SHA-256 achieves low, payload-independent latency, while RS-ID provides rigorous mathematical error bounds required for safety-critical applications. Second, a semantic synchronization policy is implemented using a physical tolerance radius R. Sensitivity analysis via heatmaps quantifies how R and verifier length n_{ver} co-influence communication speedup and verification risk. By projecting these results onto a Pareto frontier, we establish a quantitative framework for selecting operating points that balance efficiency with digital twin safety. Third, under a dynamic random-walk model, the results show that increasing R makes recovery attempts less frequent, while increasing n_{ver} reduces collision-induced misses during recovery. This separation is reflected in the distribution of Successive Recovery Intervals.

In summary, this thesis establishes a quantitative mapping between physical tolerance, quantization, and communication reliability. The results provide specific design criteria for selecting verifier lengths that ensure reliable state verification while maximizing bandwidth efficiency. These findings enable the deployment of identification-based synchronization in modern cyber-physical systems, maintaining real-time state consistency under strict communication and safety constraints.

Keywords: Digital Twins, Semantic Synchronization, Ultra-Reliable Low-Latency Communications, Networked Control Systems, Successive Recovery Intervals

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Acronyms

6G	Sixth Generation Networks	1, 4, 38
AoI	Age of Information	12
BIBO	Bounded-Input Bounded-Output	13
CDF	Cumulative Distribution Function	35, 36, 39, V
CPS	Cyber-Physical Systems	1
CR	Common Randomness	8, 9
CSI	Channel State Information	38
DRBG	Deterministic Random Bit Generator	9
DT	Digital Twin	1, 4–6, 8, 9, 11–19, 23, 24, 28, 33, 37–39, IV, V
ETC	Event-Triggered Control	5, 6, 13
GPU	Graphics Processing Unit	39
HRC	Human-Robot Collaboration	12
ID	Identification	1, 2, 4, 6, 9, 11, 12, 14, 16, 17, 20, 21, 25, 26, 28, 29, 37, IV, V
IFT	Inter-Check Interval	24
IoT	Internet of Things	38
K-ID	K-Identification	2–4, 8, 9, 11–14, 17–25, 30–33, 36–39, IV, V
LiDAR	Light Detection and Ranging	10, 37
NCS	Networked Control System	12
PLKG	Physical Layer Key Generation	9, 38
PRNG	Pseudo-Random Number Generator	9, 38
PTP	Precision Time Protocol	9, 38
RS-ID	Reed-Solomon Identification	2, 6–10, 14, 16, 18, 21, 25–27, 29, 30, 37, IV–VI
SHA	Secure Hash Algorithm	6–10, 14, 16, 18, 21, 25–27, 29–31, IV–VI
SRI	Successive Recovery Interval	2, 19, 24, 35, 36, 38, 39, V
TCP	Transmission Control Protocol	5
TDD	Time-Division Duplex	38
UDP	User Datagram Protocol	5
URLLC	Ultra-Reliable Low-Latency Communications	1, 4, 5
VoI	Value of Information	12
ZOH	Zero-Order Hold	19, 39

1 Introduction

1.1 Background and Motivation

Digital Twins (DTs) have become a standard component in modern industrial automation for connecting physical assets with their virtual counterparts. By maintaining accurate virtual copies of systems like industrial robots, DTs enable real-time monitoring, predictive maintenance, and remote control. In time-critical Cyber-Physical Systems (CPS) applications, the utility of a DT is strictly defined by the synchronization accuracy between the physical robot and the digital model [20]. Consequently, ensuring state consistency typically demands Ultra-Reliable Low-Latency Communications (URLLC) capabilities to support high-frequency updates.

However, achieving this synchronization through standard periodic transmission is often inefficient in bandwidth-constrained environments. In many control architectures, the DT acts as a global supervisor that maintains a high-fidelity state estimate. When the DT transmits full state updates to the robot at every time step, it results in significant data redundancy. Since the robot already possesses local sensors to maintain a short-term estimate, "blindly" broadcasting the entire state vector from the DT consumes scarce network resources without providing new actionable information. This mismatch between the DT's global knowledge and the robot's local perception creates an opportunity to replace continuous payload reconstruction with sparse, verification-based communication.

To mitigate this redundancy, recent research proposes identification-based Identification (ID) synchronization [12]. Unlike classical Shannon-type communication, which aims to reconstruct the full message, ID schemes focus on a verification task: confirming whether the remote state matches a local hypothesis. Building on this concept, prior works [16] demonstrate that transmitting a compressed "verifier" (e.g., a short hash or randomized code) instead of the full state can significantly reduce communication traffic. This aligns with semantic communication in Sixth Generation Networks (6G) networks [5], shifting the focus from what data is sent to what value the data holds for control.

In real-time control loops, deploying ID synchronization exposes two engineering trade-offs that motivate this thesis. First, the computational overhead of encoding on edge devices may offset the time saved by reducing transmission traffic. This shift from a communication-bound to a computation-bound bottleneck calls for an end-to-end latency analysis that accounts for both compute and communication. Second, reliability becomes a primary concern when moving from exact to semantic verification. In practice, enforcing bitwise synchronization (standard ID) can be overly conservative, triggering unnecessary updates for benign deviations such as sensor noise. This motivates semantic verification over “safety sets”, i.e., K-Identification (K-ID). However, the relaxation introduces collision risks, specifically Type II (miss) errors, where unsafe desynchronizations are incorrectly classified as admissible.

1.2 Problem Statement

Existing research on ID has primarily focused on information-theoretic capacity or simple traffic suppression, lacking a comprehensive evaluation of latency, physical safety, and computational feasibility. Specifically, it remains unclear under what bandwidth and computational constraints ID truly outperforms traditional transmission, and how to quantify the Pareto trade-off between semantic tolerance and safety risk in K-ID systems.

Moreover, semantic tolerance introduces structural constraints that are not visible in purely static formulations: as the acceptance set grows with tolerance, the finite verifier space $2^{n_{ver}}$ can become too small to distinguish all tolerated states, imposing a feasibility limit on the achievable miss probability. Characterizing these feasibility limits and their implications in dynamic settings is part of the open problem addressed in this thesis.

1.3 Contributions

To address these limitations, this thesis proposes a comprehensive framework ranging from theoretical modeling to dynamic evaluation, investigating not only the static identification problem but also error evolution in dynamic environments.

First, we develop an end-to-end latency model combining computational and transmission delays to compare algebraic coding (Reed-Solomon Identification (RS-ID)) and truncated hashing based on SHA-256. For the evaluated software stack, truncated SHA-256 achieves low and largely payload-independent encoding latency, while RS-ID incurs higher computation that depends on payload symbolization. These results quantify a practical compute–communication trade-off for edge deployments, while also motivating RS-ID for safety-critical deployments requiring rigorous mathematical guarantees (e.g., provable upper bounds under stated assumptions) on worst-case error-detection failure probability.

Second, we generalize identification to K-ID, introducing semantic tolerance to filter benign jitter. Through parameter sweeps, we map the risk–speedup Pareto frontier and show that moderate tolerance can substantially reduce fallback frequency while keeping safety risks bounded. The analysis also indicates a knee point where additional tolerance yields diminishing speedup while reducing verifier selectivity.

Third, the evaluation is extended to a time-correlated random-walk drift model to capture risks that static analysis may overlook. The results demonstrate that verifiers with insufficient bit-depth (e.g., $n_{ver} = 8$) can transform isolated collisions into persistent, undetected boundary breaches once the state drifts beyond the tolerance R. To quantify synchronization throughput, we analyze full-state recovery timing via the Successive Recovery Interval (SRI). The dynamic results separate the roles of the parameters: R mainly determines how often boundary crossings trigger recovery attempts (updates become sparser as R increases), while n_{ver} determines how often a triggered recovery is missed due to collisions.

1.4 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews related work, covering the transition from data reconstruction to semantic identification, the mathematical principles of encoding mechanisms, and the control-theoretic background. Chapter 3 details the system model, data flow design, and the derivation of theoretical error bounds for K-ID. Chapter 4 describes the implementation of the modular Python simulation framework and the experimental design strategy. Chapter 5 presents the experimental results, including latency sensitivity analysis, Pareto optimization of K-ID, and risk assessment under dynamic drift scenarios. Finally, Chapter 6 concludes the thesis and outlines future research directions.

2 Background and Related Work

This chapter reviews the transition from classical data transmission to identification-based synchronization, positioning it within the broader context of semantic communication [21] for 6G networks. It evaluates alternative synchronization strategies, such as differential encoding and event-triggered control, to motivate the engineering need for the identification (ID) approach. It then compares practical encoding mechanisms by contrasting the avalanche dynamics of cryptographic hashing with the geometric properties of algebraic coding. Finally, it discusses how this approach generalizes to K-ID, where strict equality constraints are relaxed to semantic safety sets to accommodate natural variability in physical control loops.

2.1 From Data Reconstruction to Semantic Identification

In time-critical control loops, the utility of a DT depends on its synchronization with the physical asset. Conventional protocols treat this as a data reconstruction problem, where the robot periodically transmits its full state vector S_t (e.g., joint angles and velocities) to the DT. While robust, this approach overlooks a key redundancy: the DT, using its internal kinematic model, already maintains a prediction $\hat{S}_t$ of the robot's state. The communication task is therefore not to inform the DT "where the robot is," but to verify "whether the DT's prediction is correct."

This perspective is consistent with the semantic-communication view [18] in 6G networks. Unlike classical Shannon-type communication, which emphasizes accurate reconstruction, semantic communication emphasizes information that matters for the receiver's goal. In the context of URLLC, this suggests prioritizing "actionable" information (such as safety violations) over raw telemetry.

In this setting, the coding objective shifts from Shannon's transmission capacity to the identification capacity defined by Ahlswede and Dueck [2]. In transmission, each possible state must map to a disjoint decoding set, limiting the number of distinguishable states to 2^{nR} for blocklength n. In contrast, identification permits overlaps between decoding sets, as long as pairwise intersections are statistically negligible. This relaxation enables a doubly exponential growth in identifiable states ($2^{2^{nR}}$), meaning that short verifiers can discriminate among a large state space while allowing a vanishingly small error probability.

2.2 Alternative Synchronization Strategies

Before detailing the implementation of identification, this section evaluates why established data reduction strategies (specifically Delta Encoding and Event-Triggered Control) are often insufficient for unreliable edge networks.

2.2.1 Full State Transmission (The Baseline)

The standard baseline in industrial automation is the periodic transmission of the full state vector S_t. This approach is favored for its statelessness; each packet is self-contained. If a packet at time t is lost due to channel interference, the packet at $t + 1$ allows the DT to immediately recover the correct state. However, this robustness comes at the cost of substantial bandwidth overhead. In steady-state operations, where $S_t \approx S_{t-1}$, the information entropy of the transmitted packet is near zero, yet it consumes the full bandwidth allocated to the payload n_{data}.

2.2.2 Delta Encoding and the Drift Problem

To address this redundancy, Delta Encoding [14] (or differential transmission) is frequently employed. This method transmits only the difference vector $\Delta_t = S_t - S_{t-1}$. Since Δ_t typically has a smaller dynamic range than the absolute state S_t, it can be quantized with fewer bits, offering significant compression ratios.

However, Delta Encoding introduces a key vulnerability: statefulness. The reconstruction of the current state at the receiver, denoted as S'_t, depends recursively on the history of received updates: $S'_t = S'_{t-1} + \Delta_t$. This recursive dependency creates a well-known failure mode in unreliable transport protocols like User Datagram Protocol (UDP). A single lost packet Δ_k causes a persistent offset between the physical robot and the DT, a phenomenon known as "state drift." [10] While reliable protocols like Transmission Control Protocol (TCP) can prevent packet loss, retransmissions introduce non-deterministic latency jitter, which can violate the timing deadlines of real-time control loops. Therefore, Delta Encoding poses a high risk for URLLC applications.

2.2.3 Event-Triggered Control (ETC)

Another prominent alternative is Event-Triggered Control (Event-Triggered Control (ETC)). Unlike Time-Triggered (periodic) systems, ETC only transmits data when the deviation between the current state and the last transmitted state exceeds a threshold δ: $\|S_t - S_{last\_sent}\| > \delta$ [19]. This approach effectively silences the network during periods of inactivity.

However, ETC relies fundamentally on a "push" communication model. A silent channel is ambiguous to the receiver: it could indicate that the system is operating safely within bounds, or it could indicate that the sensor has failed or the link has been severed. In safety-critical robotics, such ambiguity is often unacceptable. Furthermore, ETC requires the sensing node to perform continuous monitoring and threshold checking, pushing computational complexity to the energy-constrained edge [6].

2.2.4 The Identification Advantage

In contrast to these alternatives, the ID scheme proposed in this thesis combines statelessness with strong compression. Unlike Delta Encoding, ID verification checks the absolute state S_t directly and is therefore immune to drift; if a packet is lost, the next packet again provides an independent check. Unlike ETC, ID supports a "verify" model, allowing the DT to actively confirm synchronization with minimal overhead. This combination is well-suited for maintaining high-fidelity DTs over unreliable networks.

2.3 Encoding Mechanisms: Mathematical Structure and Bias

To realize the theoretical capacity of identification in edge computing, the encoding process must map a high-dimensional state space $\mathcal{S}$ into a low-dimensional verifier space $\mathcal{V}$ while minimizing collision probabilities. This section examines the internal mechanisms of two candidate algorithms, cryptographic hashing (Secure Hash Algorithm (SHA)-256) and algebraic coding (RS-ID), to clarify the mathematical origins of their reliability.

2.3.1 Cryptographic Construction: The SHA-256 Architecture

To understand the source of the high entropy in the generated verifiers, it is necessary to examine the internal structure of the SHA-256 algorithm [15]. SHA-256 belongs to the SHA-2 family and is built upon the Merkle-Damgård construction, which iteratively processes the input state S using a dedicated compression function. The process consists of three mathematically distinct stages: preprocessing, message expansion, and the state update loop.

Preprocessing and Padding

The input message M (comprising the state S concatenated with the seed r) is first padded to ensure its length is a multiple of 512 bits. The padding scheme appends a single '1' bit, followed by k zero bits, and finally a 64-bit integer representing the original message length. This padded message is parsed into N blocks of 512 bits, denoted as $M^{(1)}, M^{(2)}, \dots, M^{(N)}$.

Message Schedule (Expansion)

The core of the diffusion process begins with the Message Schedule. Each 512-bit block $M^{(i)}$ is expanded from 16 words (32-bit each) into 64 words, denoted as $W_0, \dots, W_{63}$. The first 16 words are copied directly from the message block. The remaining 48 words are generated recursively, creating a complex dependency chain where every future bit depends on multiple past bits:

$$W_t = \sigma_1(W_{t-2}) + W_{t-7} + \sigma_0(W_{t-15}) + W_{t-16} \pmod{2^{32}} \tag{2.1}$$

where the small sigma functions σ_0 and σ_1 introduce the initial bitwise diffusion via rotation (ROTR) and shifting (SHR):

$$\sigma_0(x) = ROTR^7(x) \oplus ROTR^{18}(x) \oplus SHR^3(x) \tag{2.2}$$

$$\sigma_1(x) = ROTR^{17}(x) \oplus ROTR^{19}(x) \oplus SHR^{10}(x) \tag{2.3}$$

This expansion ensures that a single bit flip in the input $M^{(i)}$ propagates to affect a vast majority of the 64 words in the schedule, setting the stage for the avalanche effect.

The Compression Loop

The 64 expanded words W_t are then fed into the compression function. The working state consists of eight 32-bit variables, labeled a through h, initialized to constant prime-root values. For each step t from 0 to 63, the state is updated using two temporary variables T_1 and T_2:

$$T_1 = h + \Sigma_1(e) + Ch(e, f, g) + K_t + W_t \tag{2.4}$$
$$T_2 = \Sigma_0(a) + Maj(a, b, c) \tag{2.5}$$

Here, K_t represents 64 constant 32-bit words. The non-linear logical functions Ch (Choose) and Maj (Majority), along with the big sigma functions Σ_0 and Σ_1, serve as the mixing engine:

$$Ch(x, y, z) = (x \wedge y) \oplus (\neg x \wedge z) \tag{2.6}$$
$$Maj(x, y, z) = (x \wedge y) \oplus (x \wedge z) \oplus (y \wedge z) \tag{2.7}$$
$$\Sigma_0(x) = ROTR^2(x) \oplus ROTR^{13}(x) \oplus ROTR^{22}(x) \tag{2.8}$$
$$\Sigma_1(x) = ROTR^6(x) \oplus ROTR^{11}(x) \oplus ROTR^{25}(x) \tag{2.9}$$

After calculating T_1 and T_2, the registers are shifted $(h \leftarrow g, g \leftarrow f, \dots)$, and updated as:

$$e \leftarrow d + T_1, \quad a \leftarrow T_1 + T_2 \tag{2.10}$$

This rigorous mixing guarantees that after 64 rounds, every bit of the final hash output is a complex, non-linear function of every bit of the input block. Finally, the output verifier v is obtained by truncating this digest to the target length n_{ver}, retaining the high-entropy properties generated by this procedure.

2.3.2 Algebraic Construction: The Geometry of RS-ID

In contrast to the bitwise diffusion of SHA-256, RS-ID relies on the geometric properties of polynomials over finite fields [17].

Data as a Polynomial Curve

Let the state vector S be decomposed into k segments $\{s_0, s_1, \dots, s_{k-1}\}$, where each segment is an element of the Galois Field $\mathbb{F}_q$. We interpret these segments not as raw data, but as coefficients of a unique polynomial curve $P_S(x)$ of degree $k - 1$:

$$P_S(x) = s_0 + s_1 x + s_2 x^2 + \cdots + s_{k-1} x^{k-1} = \sum_{i=0}^{k-1} s_i x^i \tag{2.11}$$

The generation of a verifier v using a random seed r is geometrically equivalent to sampling this curve at the coordinate $x = r$:

$$v = P_S(r) \pmod q \tag{2.12}$$

Collision as Intersection

A collision (Type II error) occurs when the true state S and the predicted state $\hat{S}$ are distinct ($S \neq \hat{S}$), yet their verifiers match ($v = \hat{v}$). In our geometric model, this implies:

$$P_S(r) = P_{\hat{S}}(r) \implies P_S(r) - P_{\hat{S}}(r) = 0 \tag{2.13}$$

Let $\Delta(x) = P_S(x) - P_{\hat{S}}(x)$ be the difference polynomial. Since $S \neq \hat{S}$, $\Delta(x)$ is a non-zero polynomial with degree at most $k-1$. By the Fundamental Theorem of Algebra, a non-zero polynomial of degree d can have at most d roots.

This geometric constraint provides the rigorous upper bound for the collision probability. The number of "bad seeds" r that cause a collision is limited to the number of intersection points between the two curves $P_S(x)$ and $P_{\hat{S}}(x)$, which is at most $k-1$. Thus, for a field of size q, the probability is:

$$P_{miss} = \frac{\text{Number of Roots}}{q} \leq \frac{k-1}{q} \tag{2.14}$$

This derivation visualizes the identification process as "checking for intersection". Since two distinct curves cannot intersect everywhere, checking a random point r effectively distinguishes them with high probability [17].

2.4 Common Randomness: Necessity and Generation

The transition from deterministic data compression to identification-based verification requires a fundamental shift in how mapping collisions are managed. This section discusses the necessity of common randomness (Common Randomness (CR)) in eliminating persistent synchronization failures and evaluates the practical mechanisms for its generation in edge networks.

2.4.1 Structural Blindness and the Role of Randomization

A critical limitation of any deterministic mapping from a high-dimensional state space $\mathcal{S}$ to a low-dimensional verifier space $\mathcal{V}$ is the risk of "structural blindness". In a purely deterministic scheme, specific error patterns $E_t = S_t - \hat{S}_t$ may consistently map to the same verifier collision. For example, if the mapping logic contains a periodic or modulo-based structure, a physical deviation that aligns with the mapping's kernel will remain undetected ($v = \hat{v}$) regardless of its magnitude. Even as the robot's state S_t evolves, this systematic collision can persist over time, leading to a permanent Type II (miss) error that violates the safety constraints of the DT.

To eliminate structural blindness, identification theory mandates the use of common randomness, i.e., a shared seed r that varies with each transmission. In RS-ID, varying r changes the evaluation point at which the state polynomial is sampled. Since two distinct polynomials of degree $k-1$ can agree on at most $k-1$ points, repeated sampling at different seeds prevents the same structured deviation from remaining undetected indefinitely.

In SHA-256, a time-varying salt (e.g., hashing $S_t \parallel r_t$) changes the compression input at every time step and triggers a fresh avalanche. As a result, whether a particular error pattern collides at time t does not carry over deterministically to time $t+1$.

Overall, randomization converts potential "permanent failures" into transient probabilistic events, which is a prerequisite for the reliable operation of the K-ID protocol.

2.4.2 Operational Generation of Shared Entropy

The theoretical validity of ID codes hinges on the availability of CR as a shared source of entropy between the robot and the DT. The most efficient implementation for edge deployments is the use of synchronized Pseudo-Random Number Generators (PRNGs), formally standardized as Deterministic Random Bit Generators (DRBGs) [4]. By initializing identical generators with a pre-shared secret key and tying the generation to a synchronized time index—typically maintained via the Precision Time Protocol (PTP) [9]—both nodes can locally derive an identical sequence of seeds r_t for each time slot without additional communication overhead.

However, this method introduces a critical dependency on clock synchronization. Any temporal drift between the robot and the DT results in a seed mismatch ($r_{robot} \neq r_{DT}$), causing the verifier checks to fail even when the states are physically synchronized. Such mismatches manifest as a burst of false positives, forcing the system into a bandwidth-intensive fallback mode. While alternative approaches such as Physical Layer Key Generation (PLKG) attempt to extract randomness from channel reciprocity [13], they often incur reconciliation overhead that complicates real-time control. Consequently, this thesis assumes an idealized PRNG-based synchronization model to isolate the performance of the core encoding and verification mechanisms.

2.5 Theoretical Limits and Algorithmic Trade-offs

The transition from exact identification to the K-ID framework requires a rigorous characterization of the reliability boundaries. The choice between algebraic (RS-ID) and cryptographic (SHA-256) implementations reflects a fundamental trade-off between deterministic error bounds and computational throughput.

2.5.1 Error Models and the Scaling of p_{miss}

The reliability of ID-based synchronization is governed by two distinct error models. For cryptographic hashing via SHA-256, the verifier is analyzed under the Random Oracle Model, where a truncated digest of length n_{ver} behaves as a uniform random mapping. The miss probability is statistically uniform and independent of the state dimensionality k, following $p_{miss} \approx 2^{-n_{ver}}$.

In contrast, the reliability of a single-layer RS-ID is strictly constrained by the distance properties of polynomials over the Galois field $\mathbb{F}_{2^{n_{ver}}}$. For a state vector represented by k symbols, the collision probability is upper-bounded by $\lambda_2 \leq (k-1)/2^{n_{ver}}$. This structural dependency implies that as the payload dimensionality k increases, the verifier must distinguish a larger set of potentially intersecting curves, which exposes a field-size limitation in the single-layer polynomial construction.

2.5.2 Field-Size Limitation and the Role of Concatenation

A critical bottleneck in the rudimentary polynomial model is that the size of the evaluation field $Q = 2^{n_{ver}}$ must be significantly larger than the number of state symbols k. In a single-layer construction, the miss probability is bounded by $P_{miss} \leq (k-1)/Q$. If the state dimensionality

k approaches or exceeds Q, the verifier space becomes too small, rendering it mathematically impossible to guarantee a seed r that distinguishes two distinct states via a single sample. Under such conditions, the verifier size n_{ver} must grow logarithmically with the data size ($n_{ver} > \log_2 k$) to maintain any meaningful reliability.

To overcome this limit without increasing the transmitted verifier length, we employ Concatenated RS Codes. This construction operates over a virtual extension field $GF(Q^m)$, where m represents the concatenation depth. By treating the state as a polynomial over this expanded field, the effective algebraic constraint is pushed to an exponentially larger field $Q_{eff} = Q^m$.

The suppression of the miss probability under concatenation can be demonstrated by analyzing the multi-stage evaluation process. For the m-layer construction implemented in this thesis, the provable upper bound on the collision probability λ_2 is transformed to:

$$P_{miss,RSID} \leq \frac{k_{outer} - 1}{Q^m} \tag{2.15}$$

where k_{outer} denotes the number of symbols in the outer code. Since $Q^m = (2^{n_{ver}})^m$, the denominator grows exponentially with the depth m. Even for a moderate depth of $m = 2$, the expanded field Q^2 far exceeds the typical state dimensionality k_{outer} found in robotic telemetry. When $Q^m \gg k_{outer}$, the term $(k_{outer} - 1)/Q^m$ becomes negligible, allowing the algebraic verifier's reliability to be governed primarily by the truncation length n_{ver} rather than the data size.

Consequently, the concatenated RS-ID provides a deterministic error bound that is numerically comparable to the $2^{-n_{ver}}$ statistical performance of SHA-256. This structural decoupling allows the protocol to approach the theoretical double exponential limit established by Ahlswede and Dueck, enabling the verifier length to scale as $O(\log \log k)$ rather than $O(\log k)$. This ensures that for high-dimensional sensor data, the algebraic backend maintains collision resistance equivalent to cryptographic hashing while preserving the inherent advantage of provable safety bounds.

2.5.3 Algorithmic Trade-offs: Reliability vs. Computational Efficiency

The choice between these mechanisms involves a fundamental tension. From a reliability perspective, RS-ID provides a rigorous, deterministic upper bound λ_2 on the error probability, ensuring that no pair of states exceeds a worst-case threshold regardless of the input distribution. This is essential for control loops where pathological failures must be handled analytically.

However, this mathematical guarantee incurs significant computational complexity due to the finite-field arithmetic over extension fields, which scales with data size. In contrast, SHA-256 is optimized for high throughput and provides payload-independent encoding latency. This thesis therefore utilizes RS-ID as the theoretical reference for high-reliability control signals, while using SHA-256 to demonstrate scalability for bandwidth-heavy sensor payloads such as Light Detection and Ranging (LiDAR).

2.6 Semantic Verification: Adapting to Control Flexibility

While standard identification provides a rigorous method for exact state verification, its binary nature (accepting only a single, precise value) often contradicts the operational flexibility required in modern robotic control. In many practical scenarios, the DT does not need to enforce a strict equality constraint ($x = \hat{x}$) but rather needs to guarantee that the robot operates within a permissible semantic range. This motivates K-ID, i.e., identification over a semantic acceptance set.

2.6.1 Mathematical Definition and Terminology

To align the discussion with classical identification terminology, we separate the *physical* tolerance from the *logical* acceptance-set size. Let x denote a scalar state component with prediction $\hat{x}$ and error $e = x - \hat{x}$. The physical tolerance is described by a radius $R > 0$ in the continuous state space, and deviations satisfying $|e| \leq R$ are treated as semantically admissible.

The corresponding semantic acceptance set around $\hat{x}$ is

$$\mathcal{A}_R(\hat{x}) \triangleq \{x : |x - \hat{x}| \leq R\}, \tag{2.16}$$

and its cardinality is denoted by

$$K \triangleq |\mathcal{A}_R|. \tag{2.17}$$

In a discretized state space with quantization step Δ, K is induced by R. For the 1D case,

$$K = |\mathcal{A}_R| = 2\left\lfloor \frac{R}{\Delta} \right\rfloor + 1. \tag{2.18}$$

In higher dimensions, K scales with the volume of the tolerance region at resolution Δ.

The verifier bit-length n_{ver} determines the verifier space size $2^{n_{ver}}$.

Here, K is not a physical distance; it measures the size of the "collision surface" induced by the semantic relaxation. Under the standard accept-set approximation, the per-check miss probability scales as

$$p_{miss} \approx \frac{K}{2^{n_{ver}}}. \tag{2.19}$$

When K approaches or exceeds $2^{n_{ver}}$, the verifier space is too small and reliable identification cannot be guaranteed.

The K-ID protocol suppresses *full-state recovery transmissions* as long as the physical error remains within tolerance ($|e| \leq R$). When the drift exceeds the tolerance ($|e| > R$), recovery should be triggered; a miss occurs if the drifted state falls within the K admissible states that map to an acceptable verifier outcome.

2.6.2 The Autonomous Driving Scenario

Consider a practical example of an autonomous robot (or vehicle) navigating a public roadway. The control policy might specify a safe velocity corridor, such as "maintain speed between 40 km/h and 60 km/h." If the robot moves at 45 km/h while the DT predicted 50 km/h, a standard ID scheme would flag this as a collision (mismatch) because the hashed values of "45" and "50" are distinct. This triggers an unnecessary data transmission to correct a behavior that is semantically valid.

K-ID resolves this inefficiency by defining the acceptance region $\mathcal{B}$ as the interval $[40, 60]$. As long as the robot's state falls within this set, the K-ID verifier confirms validity without consuming bandwidth for minor deviations.

2.6.3 Human-Robot Collaboration and Jitter

This logic extends beyond simple velocity constraints to more complex spatial and interactive scenarios. For instance, in Human-Robot Collaboration (HRC), the exact coordinates of a robotic arm are less critical than ensuring the end-effector remains outside a defined "human safety bubble" [20]. Similarly, for high-precision assembly tasks involving compliant mechanisms, the robot may exhibit minor high-frequency vibrations (jitter) that are mechanically acceptable. Standard ID would treat these micro-vibrations as errors, whereas K-ID can be tuned to ignore high-frequency noise while still detecting significant trajectory deviations.

By shifting the verification criterion from data exactness to semantic compliance, K-ID aligns the synchronization protocol with the actual control objectives. It allows the system to filter out benign variability, including speed fluctuations, sensor noise, and compliant jitter, and to focus limited communication resources on detecting genuine safety violations. In this thesis, we implement this semantic verification using randomized encoding under edge constraints.

2.6.4 Discussion: Trading Freshness for Value

The scenarios described above illustrate a deviation from traditional network optimization. Conventional real-time protocols often aim to minimize the Age of Information (AoI) [11], keeping the DT's estimate as fresh as possible regardless of content. Under an AoI-minimization strategy, the robot would transmit its velocity continuously even when cruising steadily within the safe corridor, purely to prevent the timestamp from "aging."

K-ID challenges this baseline by introducing a semantic filter that prioritizes the Value of Information (VoI) over simple freshness. As demonstrated in the autonomous driving case, a "stale" position estimate (high AoI) that remains within the safety tolerance $\mathcal{B}$ retains its utility for the control loop. By allowing AoI to increase during semantically safe intervals, K-ID trades unnecessary freshness for bandwidth efficiency. This is consistent with goal-oriented communication [18], where the objective shifts from minimizing average latency to minimizing the latency of meaningful (unsafe) updates.

2.7 Control-Theoretic Perspectives: Deadbands and Stability

Beyond communication efficiency, the deployment of K-ID changes the control loop structure. From a control-theoretic perspective, K-ID yields a Networked Control System (NCS) operating under a semantic sampling policy. This section analyzes the implications for stability and execution semantics, and maps K-ID parameters to established control concepts.

2.7.1 K-ID as Deadband Control

The tolerance mechanism in K-ID, defined by the acceptance set $\mathcal{B}$, is mathematically equivalent to Deadband Control (or Dead-Zone Control) in non-linear systems theory. In classical mechanics, deadbands are often viewed as parasitic nonlinearities (e.g., gear backlash) that degrade performance. However, in resource-constrained NCS, artificially introduced deadbands are a recognized strategy for reducing actuator wear and communication load [3].

By suppressing full-state updates when the synchronization error $e_t = S_t - \hat{S}_t$ satisfies $|e_t| \leq R$, K-ID implements Lebesgue Sampling (sampling based on value domains), as opposed to the traditional Riemann Sampling (sampling based on time domains). This semantic filtering prevents the control loop from reacting to high-frequency, low-amplitude noise (chattering), thereby reducing mechanical stress on physical actuators while preserving bandwidth for significant trajectory corrections.

2.7.2 Prevention of Zeno Behavior

A known challenge in standard Event-Triggered Control (ETC) is the risk of Zeno behavior (a phenomenon where the inter-event time converges to zero, causing the system to trigger an infinite number of updates in a finite time interval) [6]. This can occur when the state trajectory "slides" along the triggering threshold, potentially leading to computational livelock or bursts of network traffic.

K-ID avoids this instability through its hybrid architecture. Although the decision to recover is event-driven (based on the semantic check $v \in \mathcal{V}_R$, derived from $\mathcal{A}_R$), the verification opportunity occurs at discrete, time-triggered intervals (ticks). This imposes a strict lower bound on the inter-transmission time, denoted as $T_{min} \geq T_{tick}$. By design, K-ID guarantees Zeno-freeness, ensuring that the network load remains deterministically bounded even in worst-case divergence scenarios.

2.7.3 Bounded-Input Bounded-Output (BIBO) Stability

The introduction of the semantic tolerance radius R raises the question of closed-loop stability. Under K-ID, the DT (controller) operates on an estimated state $\hat{S}_t$ that deviates from the true plant state S_t by an error term δ_t. Assuming the absence of collision-induced miss errors (which are probabilistically negligible as shown in Section 2.3), the synchronization error is strictly bounded:

$$||\delta_t|| = ||S_t - \hat{S}_t|| \leq R \tag{2.20}$$

From a stability analysis perspective, this bounded error acts as a bounded disturbance input to the controller. According to the principle of Bounded-Input Bounded-Output (BIBO) stability [8], as long as the underlying control law can reject disturbances of magnitude R, the closed-loop system remains stable. This implies that the choice of R is not only a traffic parameter but also a stability constraint: it must remain within the controller's stability margin (or region of attraction).

3 Design Methodology

This chapter formulates the system model for state synchronization, moving from classical transmission to identification (ID) and tolerance-based K-ID. It introduces an analytical end-to-end latency model and analyzes key engineering trade-offs, including quantization resolution, computational cost (RS-ID vs. SHA-256), and collision risk as a function of the acceptance-set size K. The chapter also introduces a time-correlated random-walk model to study safety boundaries and error accumulation in time-varying systems.

3.1 System Model and Data Flow

A networked control system is considered in which a robot (sender) synchronizes its state S_t with a DT (receiver) maintaining a prediction $\hat{S}_t$. The fundamental difference between the two approaches lies in their data processing pipelines.

3.1.1 Baseline: Traditional Transmission Flow

The standard synchronization protocol follows a linear, "always-on" approach. As illustrated in Figure 3.1, the robot first samples its sensors to generate the full state message S_t (size n_{data}) and transmits the entire payload over the channel. The DT receives the message, overwrites its local prediction $\hat{S}_t$ with the received S_t, and confirms the update.

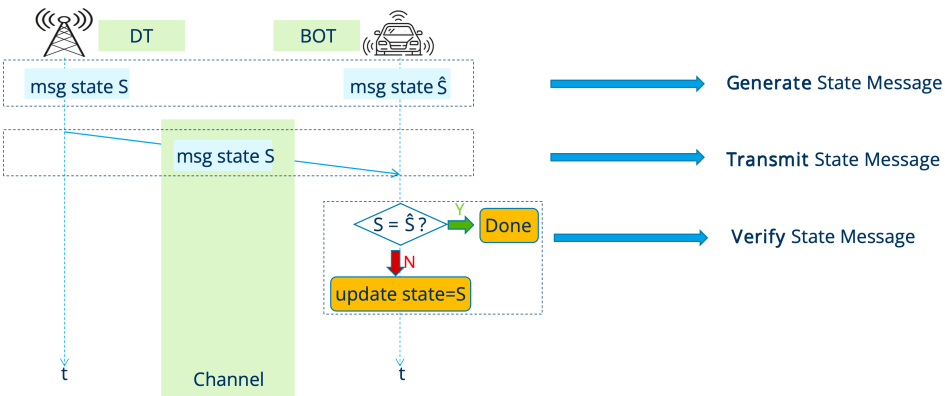


Figure 3.1: Data flow of Traditional Synchronization: The full state is transmitted regardless of the DT's knowledge.

This process is robust but inefficient, since the transmission cost is paid unconditionally. Under this baseline, the transmission latency is deterministic and scales directly with the state size, $T_{trans} = n_{data}/B$.

3.1.2 Identification (ID) Data Flow

The identification-based protocol reduces communication load by verifying consistency before sending the full state. As shown in Figure 3.2, both the robot and the DT first generate their respective state views, S_t and $\hat{S}_t$. The robot computes a compact verifier $v = E(S_t, r)$ using a shared seed r, while the DT computes the corresponding reference $\hat{v} = E(\hat{S}_t, r)$ locally. Only the short verifier v (size $n_{ver} \ll n_{data}$) is transmitted. The DT then compares v and $\hat{v}$: if they match, the protocol terminates; if they differ, the DT triggers fallback and requests the full payload S_t to resynchronize.

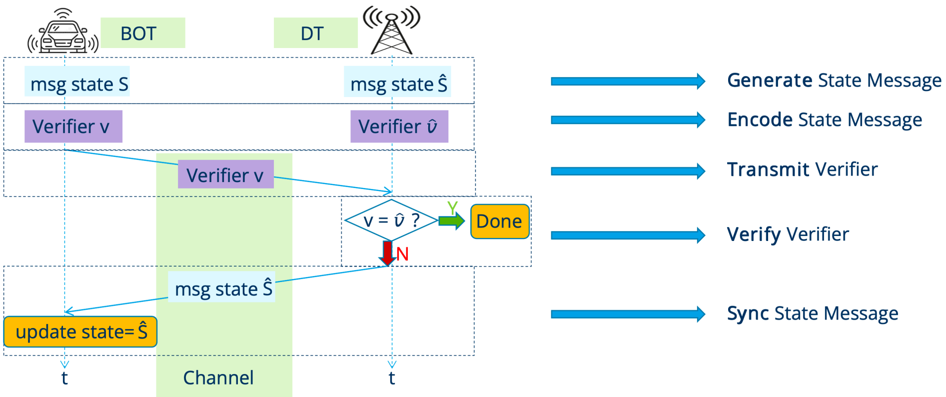


Figure 3.2: Data flow of ID Synchronization: High-bandwidth transmission is replaced by local encoding and conditional fallback.

3.1.3 Field-Size Limitation and the Role of Concatenation

A critical bottleneck in the rudimentary polynomial model is that the size of the evaluation field $Q = 2^{n_{ver}}$ must be significantly larger than the number of state symbols k. In a single-layer construction, the miss probability is bounded by $P_{miss} \leq (k-1)/Q$. If the state dimensionality k approaches or exceeds Q, the verifier space becomes too small, rendering it mathematically impossible to guarantee a seed r that distinguishes two distinct states via a single sample. Under such conditions, the verifier size n_{ver} must grow logarithmically with the data size ($n_{ver} > \log_2 k$) to maintain any meaningful reliability.

To overcome this limit without increasing the transmitted verifier length, concatenated RS codes are used. This construction operates over a virtual extension field $GF(Q^m)$, where m represents the concatenation depth. By treating the state as a polynomial over this expanded field, the effective algebraic constraint is pushed to an exponentially larger field $Q_{eff} = Q^m$.

The suppression of the miss probability under concatenation can be demonstrated by analyzing the multi-stage evaluation process. For the m-layer construction implemented in this thesis, the provable upper bound on the collision probability λ_2 is transformed to:

$$P_{miss,RSID} \leq \frac{k_{outer} - 1}{Q^m} \tag{3.1}$$

where k_{outer} denotes the number of symbols in the outer code. Since $Q^m = (2^{n_{ver}})^m$, the denominator grows exponentially with the depth m. Even for a moderate depth of $m = 2$, the expanded field Q^2 far exceeds the typical state dimensionality k_{outer} found in robotic telemetry. When $Q^m \gg k_{outer}$, the term $(k_{outer} - 1)/Q^m$ becomes negligible, allowing the algebraic verifier's reliability to be governed primarily by the truncation length n_{ver} rather than the data size.

Consequently, concatenated RS-ID provides a deterministic error bound that is numerically comparable to the $2^{-n_{ver}}$ statistical performance of SHA-256. This structural decoupling allows the protocol to approach the theoretical double exponential limit established by Ahlswede and Dueck, enabling the verifier length to scale as $O(\log\log k)$ rather than $O(\log k)$. For high-dimensional sensor data, this yields collision resistance comparable to cryptographic hashing while preserving the advantage of provable safety bounds.

3.2 End-to-End Latency Model

Based on the data flows defined in Section 3.1, the expected latency is formulated as follows.

Since the traditional flow always executes the full transmission, its latency is deterministic:

$$T_{trad} = \frac{n_{data}}{B} \tag{3.2}$$

where B is the channel bandwidth.

In contrast, the ID flow is probabilistic. The total time includes the mandatory encoding and verifier transmission, plus the conditional penalty for fallback. Matching the stages in Figure 3.2:

$$T_{ID} = \underbrace{t_{ver}}_{\text{Encode}} + \underbrace{\frac{n_{ver}}{B}}_{\text{Transmit Verifier}} + \underbrace{p_{desync}(1 - p_{miss})\frac{n_{data}}{B}}_{\text{Sync (Fallback Penalty)}} \tag{3.3}$$

Here, t_{ver} denotes the computation time of the Encode stage, p_{desync} is the probability that the DT prediction fails ($S_t \neq \hat{S}_t$), and the fallback penalty $(1 - p_{miss})\frac{n_{data}}{B}$ is incurred only when a desynchronization exists and is successfully detected by the verifier.

The model highlights the key trade-off: ID reduces the transmission term from $\frac{n_{data}}{B}$ to $\frac{n_{ver}}{B}$ but adds a computational cost t_{ver}. Therefore, ID is advantageous when the bandwidth B is limited (making transmission expensive) or when the prediction accuracy is high (keeping p_{desync} small).

3.3 Tolerance-Based Verification (K-ID)

The ID data flow described in Section 3.1 assumes a strict equality check between a single predicted state $\hat{S}_t$ and the true state S_t. In practice, many cyber-physical systems tolerate bounded deviations without requiring a full resynchronization. The verification stage is therefore extended to a tolerance-based rule, widely referred to as K-ID in the information-theory literature [1].

Figure 3.3 illustrates the resulting protocol. The operational stages mirror the standard ID flow, but the decision criterion at the receiver is generalized: instead of checking equality against a single predicted verifier $\hat{v}$, the DT accepts if the received verifier matches any candidate in a precomputed set corresponding to the tolerable neighborhood of $\hat{S}_t$.

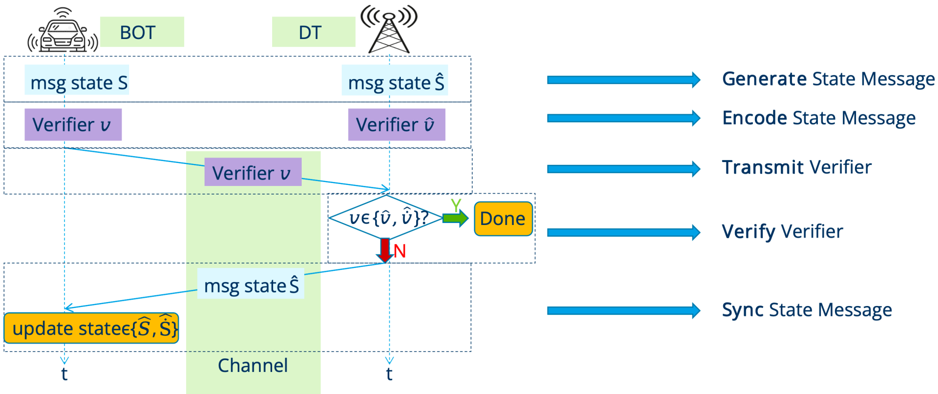


Figure 3.3: Data flow of K-ID Synchronization: High-bandwidth transmission is replaced by local encoding and conditional fallback.

3.4 System Model of Semantic Identification

3.4.1 Acceptance Set Mapping and Cardinality

The K-ID protocol operationalizes semantic synchronization by mapping a physical tolerance radius R into a discrete hypothesis set. Let $\mathcal{Q}$ denote the domain of quantized states achievable by the robot, characterized by a discretization resolution Δ. For a given DT prediction $\hat{S}_t$, the semantic acceptance set $\mathcal{A}_R$ is defined as the collection of all quantized states $s \in \mathcal{Q}$ such that $\|s - \hat{S}_t\| \leq R$.

Consistent with the terminology established in Section 2.6.1, the cardinality of this set, $K = |\mathcal{A}_R|$, represents the message set size that the identification channel must accommodate. For the one-dimensional state representation used in our experiments, this mapping follows:

$$K \approx \frac{2R}{\Delta} + 1. \tag{3.4}$$

This relationship links the physical control requirements (R, Δ) to the theoretical reliability limits of the verifier through the induced acceptance-set size K.

3.4.2 Verification and Fallback Logic

The synchronization process is executed as a two-stage verification task. At each time step, the Robot computes a single verifier $v = Enc(S_t, r)$, where r is the shared seed derived from the common randomness source described in Section 2.4. Simultaneously, the DT pre-computes a set of valid verifiers $\mathcal{V}_{set} = \{Enc(s, r) \mid s \in \mathcal{A}_R\}$ corresponding to all candidates within the acceptance set.

The DT confirms synchronization if $v \in \mathcal{V}_{set}$. This condition implies that the robot's state, while potentially deviating from the point prediction $\hat{S}_t$, remains within the semantically acceptable boundary R. If the received verifier is not present in $\mathcal{V}_{set}$, the DT identifies a safety violation and triggers a full-state fallback transmission to resynchronize the model.

3.4.3 K-ID Reliability and Latency Trade-offs

Based on the theoretical bounds derived in Section 2.5.1, the miss probability for K-ID converges for both SHA-256 and concatenated RS-ID backends. By applying the union bound over the K candidates, the system-level reliability follows:

$$p_{miss}(R, n_{ver}) \approx \frac{K}{2^{n_{ver}}}. \tag{3.5}$$

This implies that the safety risk scales linearly with the acceptance-set size K, emphasizing the importance of the verifier-space constraint ($K < 2^{n_{ver}}$) to avoid deterministic synchronization failure.

To relate these parameters to end-to-end performance, the expected latency T_{KID} is defined as a function of the tolerance radius R. Assuming the prediction error follows a Gaussian distribution $\mathcal{N}(0, \sigma^2)$, the probability of an unsafe deviation is $p_{unsafe}(R) = \Pr(|E_t| > R)$. The total latency is then modeled as:

$$T_{KID} = t_{enc} + t_{dec}(K) + \frac{n_{ver}}{B} + p_{unsafe}(R)(1 - p_{miss})\frac{n_{data}}{B}, \tag{3.6}$$

where $t_{dec}(K)$ represents the decoding complexity, which scales linearly with the set size K. The expression highlights the core optimization problem: increasing R reduces the frequency of bandwidth-heavy fallbacks ($p_{unsafe} \cdot n_{data}/B$) but increases both the computational overhead $t_{dec}(K)$ and the collision risk p_{miss}.

3.4.4 Quantization and Computational Scalability

The choice of resolution Δ dictates the granularity of the safety zone and the resulting computational workload. For a fixed physical radius R, driving $\Delta \to 0$ leads to a hyperbolic increase in K. Since the DT must perform K verifier evaluations per tick, the decoder latency $t_{dec}(K)$ can quickly saturate the real-time processing budget of edge devices. Thus, Δ is treated as a constraint satisfaction variable: it must be sufficiently fine to meet control fidelity requirements while remaining large enough to ensure $t_{dec}(K)$ fits within the sampling interval T_{tick}.

3.5 Dynamic Evaluation Model: Random Walk with Drift

While the static parameter sweeps presented in Section 5.1.2 estimate instantaneous miss probabilities, they do not capture temporal correlation in continuous control loops. To study the accumulation of unsynchronized drift and resulting boundary breaches, a time-correlated dynamic model is introduced. In this framework, the physical robot drifts stochastically while the DT maintains a local state estimate via Zero-Order Hold (ZOH) prediction.

The scalar state evolution of the robot is modeled as a discrete-time random walk:

$$S_t = S_{t-1} + \delta_t, \tag{3.7}$$

where δ_t represents the process noise. Two noise profiles are considered: a uniform increment $\delta_t \sim \mathcal{U}(-\Delta_{step}, \Delta_{step})$ representing constrained kinematic bounds, and a Gaussian increment $\delta_t \sim \mathcal{N}(0, \sigma^2_{step})$ representing unbounded disturbances.

On the receiver side, the DT assumes state consistency in the absence of explicit updates. Under the ZOH assumption, the prediction remains constant between corrections:

$$\hat{S}_t = \hat{S}_{t-1}, \quad \text{for } t \in [t_k, t_{k+1}), \tag{3.8}$$

where t_k is the timestamp of the last successful full-state synchronization. Consequently, the instantaneous synchronization error $E_t = |S_t - \hat{S}_t|$ diverges stochastically as the physical system drifts away from the frozen digital estimate.

At each discrete simulation tick, the protocol enforces the K-ID semantic gate based on the physical tolerance R. The transmission logic is defined as:

$$\begin{cases} |E_t| \leq R: & \text{Silence (masked drift)}, \\ |E_t| > R: & \text{Verifier Transmission and Verification.} \end{cases} \tag{3.9}$$

When $|E_t| > R$, the robot transmits a verifier v_t. The DT validates v_t against the acceptance set $\mathcal{A}_R$ centered at $\hat{S}_t$. If a collision occurs (Type II error), the system suffers a *safety miss*: the DT erroneously accepts the outdated state $\hat{S}_t$, allowing the unsynchronized drift to persist and potentially escalate. Conversely, a successful verification triggers a fallback, resetting the estimate $\hat{S}_t \leftarrow S_t$ and restoring the synchronization error to zero.

To align this dynamic simulation with the reliability analysis established in Chapter 2, the probability of collision during verification events is governed by the verifier-space constraint. As established in Equation 3.5, the miss probability is approximated by:

$$p_{miss}(R, n_{ver}) \approx \frac{K}{2^{n_{ver}}}, \tag{3.10}$$

where $K = |\mathcal{A}_R| \approx 2R/\Delta + 1$ denotes the cardinality of the semantic hypothesis set. This dynamic model allows for the evaluation of the SRI, characterizing how K-ID transforms continuous physical motion into sporadic, bursty communication traffic while managing the risk of undetected boundary breaches.

4 Implementation

This chapter summarizes the practical implementation used to instantiate and evaluate the theoretical models introduced in Chapter 3. The evaluation is organized around three complementary models. First, the end-to-end ID latency model is instantiated to generate bandwidth and desynchronization sensitivity plots. Second, the tolerance-based K-ID protocol is evaluated by sweeping the physical tolerance radius R and verifier length n_{ver}. For each (R, Δ) pair, the induced acceptance-set cardinality $K = |\mathcal{A}_R|$ follows the definition in Equation (2.18). Third, a time-correlated dynamic model is implemented in which K-ID operates under random-walk drift, allowing accumulation effects such as sustained boundary breaches after missed corrections to be studied.

4.1 Data Packet Structure Abstraction

While the analysis in Chapter 3 models transmission cost in terms of n_{ver} and n_{data}, practical deployments encapsulate these fields in a communication frame. Figure 4.1 illustrates the abstract frame layout assumed in this thesis. In normal operation, the robot transmits only a compact verifier; when verification fails, the protocol falls back to transmitting the full state payload. The K-ID protocol uses the same message structure as ID; it differs only in the receiver-side acceptance rule (set membership rather than strict equality).

In the latency model, the header, timestamp, and checksum/CRC are treated as fixed overheads and are omitted for clarity. If required, these can be incorporated by adding a constant number of bits to both the verifier-only and fallback frames.

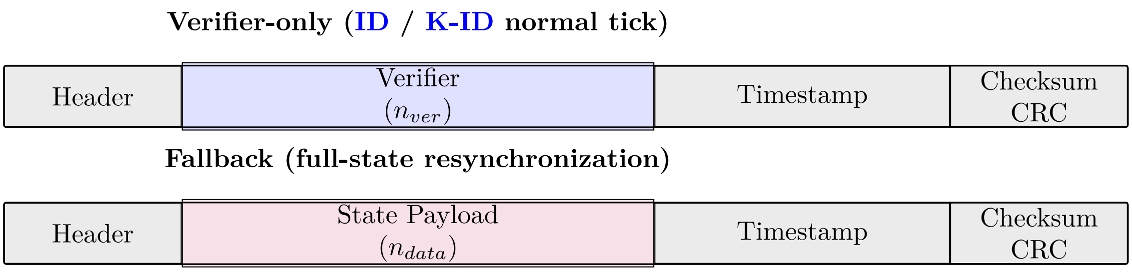


Figure 4.1: Abstract ID message format used for modeling. The normal path transmits only a verifier, while the fallback path transmits the full state payload. Fixed overhead fields are shown for completeness but are not explicitly modeled in the latency equations.

4.2 Software Architecture

A modular Python simulator supports rapid prototyping and evaluation of the proposed models.

Figure 4.2 shows the high-level decomposition used throughout the Python experiments. The simulator is structured around three interchangeable components—Encoder, Channel/Dynamics, and Decoder/Decision. This separation allows swapping SHA-256 and RS-ID encoders, switching between strict ID and tolerance-based K-ID decision rules, and evaluating both i.i.d. and time-correlated models without entangling implementation details.

This modularity enables controlled comparisons: the experiment runner, channel models, and measurement pipeline remain fixed while only the encoder backend (SHA-256 vs. RS-ID) and the receiver-side acceptance rule (ID vs. K-ID) are varied.

The interface between Encoder and Channel is a transmitted packet/frame. In normal operation this contains only a verifier tag; on fallback it contains the full state payload. This matches the message abstraction in Figure 4.1.

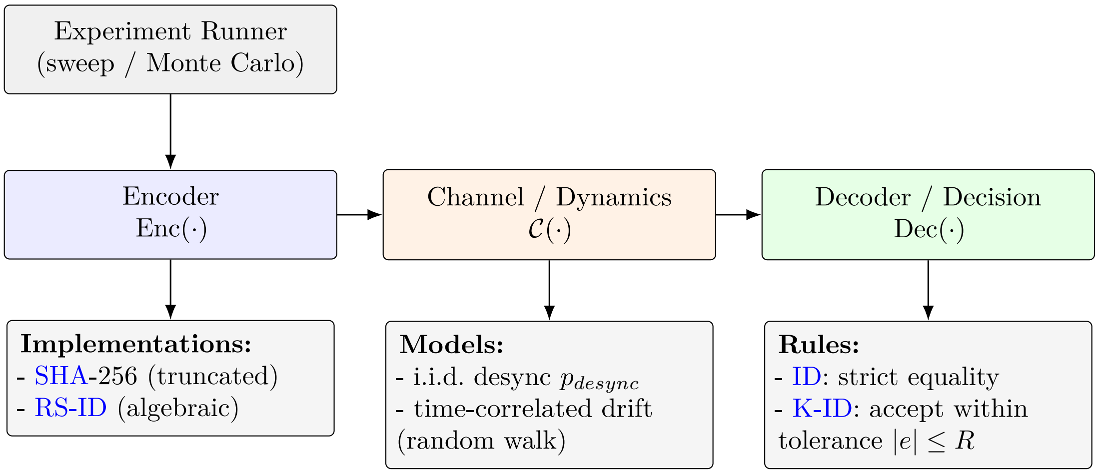


Figure 4.2: High-level simulator architecture used in the Python experiments. Encoder, Channel/Dynamics, and Decoder are decoupled so that coding schemes (SHA-256 vs. RS-ID), acceptance rules (ID vs. K-ID), and stochastic models (i.i.d. vs. time-correlated drift) can be varied independently.

4.3 K-ID Simulation Setup and Implementation

This section outlines the simulation environment used to instantiate the K-ID framework from Chapter 3. The objective is to quantify the trade-off between the physical tolerance radius R and safety risks under constrained edge-link conditions. In the simulator, R is the physical threshold in the state space, while K denotes the induced acceptance-set size $K = |\mathcal{A}_R|$ used in the collision analysis (Equation (2.19)).

4.3.1 Error Models and State Generation

The simulation models the robot's state as a continuous random variable constrained to a physical domain of 0 to 100. To evaluate the impact of error distribution shapes, two contrasting noise models are instantiated while controlling for their noise energy.

The baseline Gaussian model is centered at the domain midpoint with a standard deviation of $\sigma = 16.0$. This parameter choice maximizes utilization of the physical state space without violating boundaries, ensuring that the simulation includes rare but safety-relevant tail events. For the comparative Uniform model, a variance-matching constraint is applied to isolate the effect of bounded support. For a symmetric uniform distribution on $[-a, a]$, $\mathrm{Var} = a^2/3$; equating $a^2/3 = \sigma^2$ yields $a = \sqrt{3}\,\sigma \approx 27.71$.

This half-width defines a *bounded-support threshold*: choosing $R \geq a$ guarantees $|e| \leq R$ for all samples under the Uniform model and therefore makes tolerance violations impossible by construction.

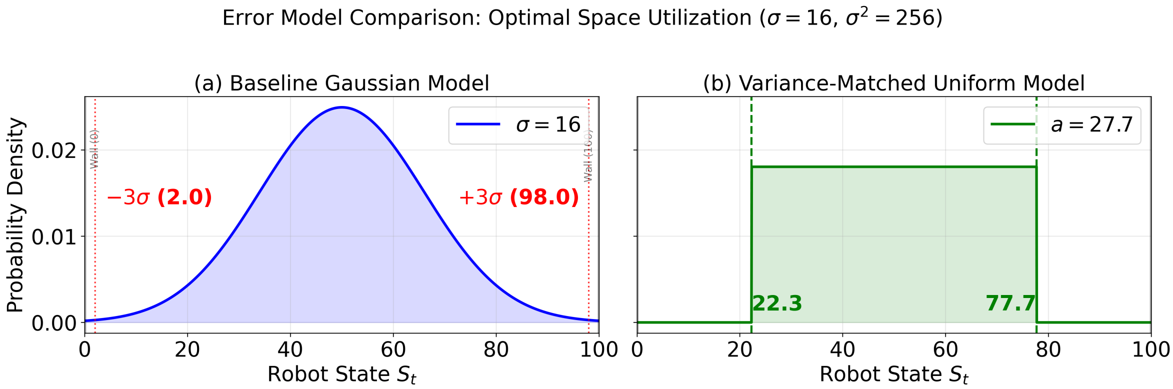


Figure 4.3: Visual comparison of the variance-matched error models configured with $\sigma = 16$. The Gaussian baseline (a) maximizes physical space utilization within the domain bounds $[2, 98]$, preventing truncation artifacts while maintaining high noise energy. The Uniform model (b) illustrates the bounded-support threshold: the selected tolerance radius of 30 exceeds the noise support, guaranteeing zero tolerance-violation risk under the Uniform model, whereas the Gaussian model retains residual tail risk outside the tolerance zone.

4.3.2 Computational Scalability and Latency Model

A sensitivity analysis quantifies the computational scalability of the K-ID decoder by sweeping the quantization resolution Δ from 1.0 down to 0.01. The total decoder latency is modeled linearly as the product of the acceptance-set cardinality K and the unit verification overhead; since K grows approximately as $K \propto R/\Delta$ (Equation (2.18)), finer quantization increases computation even when R is fixed.

The results of this sensitivity sweep reveal a scalability bottleneck governed by the resolution. As the quantization becomes finer, the computational cost grows hyperbolically. For large-tolerance configurations, high-precision settings drive the decoder latency well beyond typical control budgets. Relaxing the resolution to 0.1 reduces the worst-case latency to approximately

1.3 ms, balancing precision with feasibility. Consequently, this operating point is selected as the fixed resolution for subsequent experiments.

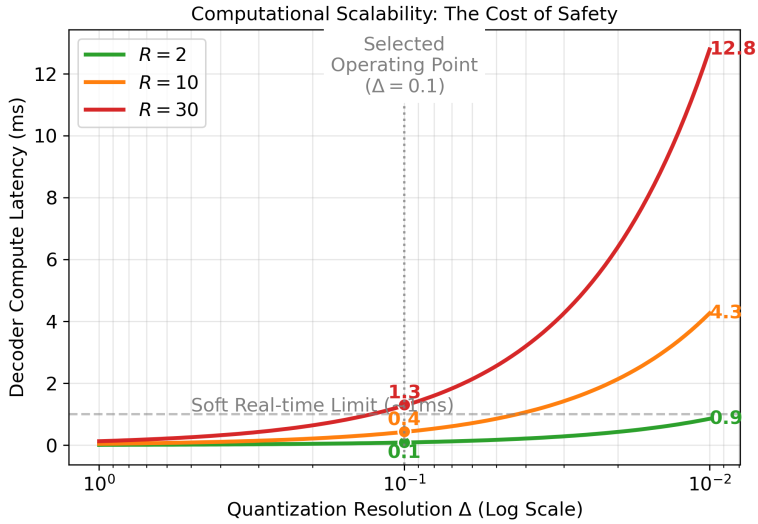


Figure 4.4: Computational scalability analysis showing decoder latency as a function of quantization resolution Δ. The red curve represents the bounded-support configuration ($R = 30$). At the high-precision limit ($\Delta = 0.01$), latency reaches a prohibitive 12.8 ms. However, at the selected operating point ($\Delta = 0.1$, indicated by the vertical dotted line), the latency drops to approximately 1.3 ms, balancing physical coverage with real-time computational feasibility.

4.3.3 System Configuration and Visualization

The primary simulation assumes a payload of 4001 bits transmitted over a 1 Mbps link. The evaluation framework employs the physical tolerance radius R as the primary independent variable to maintain a direct mapping to robotic safety requirements. The sweep uses $R \in \{2, 5, 10, 20, 30\}$. The largest value is motivated by the Uniform half-width derived above ($a \approx 27.71$), which provides a boundary condition where tolerance violations are impossible under bounded noise.

To analyze the trade-off space, a dual visualization strategy is employed. Raw simulation results are mapped onto a grid defined by verifier length and physical tolerance to display speedup factors and risk probabilities. These points are subsequently projected onto a risk-vs-speedup scatter plot to extract the Pareto frontier.

4.4 Dynamic Evaluation Framework: The Random Walk Model

To evaluate K-ID under continuous operation, a discrete-time simulation framework is used comprising a stochastic robot agent, a bandwidth-constrained channel, and a DT receiver. Unlike the static sweeps in Section 4.3, this dynamic evaluation models time-correlated accumulation of drift, allowing safety risks that only emerge over extended trajectories to be quantified.

4.4.1 Dynamic Protocol Logic

The robot's state evolution is modeled as a diffusive process, specifically a Gaussian random walk defined by $S_t = S_{t-1} + \delta_t$, where the increment follows a normal distribution with a unit step deviation. This standardized diffusion ensures the variance of the uncorrected state grows linearly with time, providing a reproducible challenge for the synchronization protocol.

The K-ID protocol operates as a guardrail mechanism on this state evolution. At each time tick, the absolute deviation between the true state and the DT estimate is evaluated. If the deviation remains within the tolerance radius ($|e| \leq R$), the protocol remains silent; if $|e| > R$, a verification event is triggered.

For long-horizon simulation, the encoder backend is abstracted as a probabilistic verification model grounded in the Chapter 2 reliability law (Equation (2.19)). Specifically, the verification outcome at a check event is implemented as a Bernoulli trial with miss probability $p_{miss} \approx K/2^{n_{ver}}$, where $K = |\mathcal{A}_R|$ is induced by (R, Δ) (Equation (2.18)). This abstraction preserves the theoretically derived dependence on (K, n_{ver}) while decoupling the experiment logic from implementation-specific hashing costs.

4.4.2 Experimental Strategy

The evaluation strategy consists of two complementary phases. First, a qualitative stress test is used to induce observable failure modes. By intentionally reducing n_{ver}, the collision probability increases, which amplifies otherwise rare long-tail events and makes sustained boundary breaches observable within a manageable simulation window.

Second, a quantitative parameter sweep systematically varies the tolerance radius and verifier length while fixing the drift dynamics. This grid search decouples the masking effect of physical tolerance from the collision risk of the verification step, providing the statistical basis for the trade-off analysis.

4.4.3 Metrics for Dynamic Safety

To quantify the trade-off between efficiency and risk, three primary metrics are defined. The Successive Recovery Interval (SRI) measures the number of ticks between two consecutive recovery-triggered check events (i.e., transmission attempts triggered by $|E_t| > R$). In the simulator, this quantity is equivalent to the inter-check interval (Inter-Check Interval (IFT)); a larger SRI indicates higher masking efficiency and reduced channel contention.

Safety is assessed via two complementary metrics. The Breach Duration counts the number of consecutive ticks during which the system remains in an unsafe state due to hash collisions, measuring the temporal persistence of safety violations. Finally, the Maximum Drift metric records the peak absolute error reached during a simulation run. Unlike Breach Duration, this quantifies the worst-case physical magnitude of the error, serving as a proxy for potential physical damage in control scenarios.

5 Results and Evaluation

This chapter presents the empirical evaluation of the proposed ID-based synchronization protocols. It first characterizes the system's fundamental parameters to define the bandwidth and state synchronization boundaries where ID yields a latency advantage over traditional transmission. Next, it reports a parameter sweep of the K-ID protocol, quantifying the safety-performance trade-offs and identifying Pareto-optimal configurations for deployment. Finally, it evaluates a time-correlated random-walk drift model to highlight risk accumulation in continuous operation.

5.1 Empirical Parameter Verification

To instantiate the latency models derived in Chapter 3, an experimental characterization of the computational cost (t_{ver}) and the theoretical reliability limits of the encoding mechanisms was performed.

5.1.1 Encoding Latency Analysis

The computational overhead was evaluated through a parameter sweep over verifier length $n_{ver} \in \{4, 8, 12, 16\}$ and payload size $n_{data} \in \{96, 4001\}$ bits in Figure 5.1. Each data point represents the average encoding time over 1000 trials, measured on a MacBook Air (2020) host system, equipped with an Apple M1 chip (8-core CPU) and 8 GB of unified memory. This platform operates on the ARMv8 architecture. All measurements were recorded under macOS 26.2 using both the optimized and reference backends for SHA-256, and the reference implementation for RS-ID.

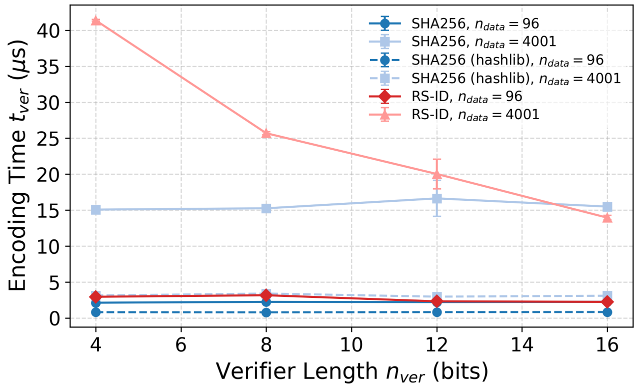


Figure 5.1: Comparative overhead analysis of ID mechanisms. For SHA-256, results are reported for both the optimized hashlib [12] and the reference high-level backend.

The measurements show that SHA-256 maintains a nearly constant latency profile across all verifier lengths for a fixed payload size. This behavior is consistent with the Merkle-Damgård construction of the SHA-256 algorithm, which processes data in fixed 512-bit blocks. As the number of iterations depends solely on the input data volume and padding rather than the final truncation length, the computational burden remains invariant to n_{ver}.

In contrast, RS-ID shows improved performance as the verifier length increases. This trend is substantiated by a detailed processing analysis presented in Table 5.1. The computational cost of RS-ID is governed by the total number of finite field symbols k required to represent the payload. According to the RS-ID construction principles, each symbol's capacity scales with n_{ver}, meaning that for a fixed payload size, an increase in verifier length leads to a substantial reduction in the total symbol count [12].

As evidenced by the measurements for a 4001-bit payload, the symbol count k decreases from 501 to 126 as n_{ver} scales from 4 to 16 bits. This reduction in the total number of operations effectively offsets the marginal increase in the computational complexity per symbol associated with larger Galois Fields. Consequently, for algebraic identification, higher precision configurations not only enhance reliability but also yield a net gain in execution efficiency.

Table 5.1: Breakdown of processing costs for block-based (SHA-256) and symbol-based (RS-ID) encoding.

		SHA-256			RS-ID		
n_{ver}	Payload	t_{ver} (μs)	Per-blk (μs)	Blocks	t_{ver} (μs)	Per-sym (μs)	k
4	96	2.06	2.06	1	2.96	0.247	12
4	4001	14.77	1.85	8	41.21	0.082	501
8	96	2.10	2.10	1	2.61	0.435	6
8	4001	14.90	1.86	8	24.87	0.099	251
12	96	2.17	2.17	1	2.36	0.589	4
12	4001	14.97	1.87	8	17.00	0.102	167
16	96	2.17	2.17	1	2.23	0.744	3
16	4001	14.92	1.87	8	13.44	0.107	126

A performance gap is observed between different implementation backends of SHA-256. The optimized implementation, which uses the OpenSSL engine and direct byte-stream processing, achieves lower latency compared to the reference backend. In high-level languages such as Python, the reference implementation is subject to substantial serialization overhead when converting integer lists into the internal C structures required for hashing. This data handling cost dominates the total latency for larger payloads, suggesting that the efficiency of data passing between software layers can be as important as the algorithmic complexity itself.

5.1.2 Theoretical Reliability Characterization

Following the theoretical framework established in Section 2.3, we assume a unified reliability model for both encoding mechanisms. As derived from the distance properties of RS-ID and the statistical behavior of SHA-256, the miss probability is structurally governed by the verifier length n_{ver}. Consequently, we apply the scaling law defined in Equation (2.19) to characterize the expected collision rates:

$$p_{miss} \approx \frac{K}{2^{n_{ver}}}, \quad K = 1 \tag{5.1}$$

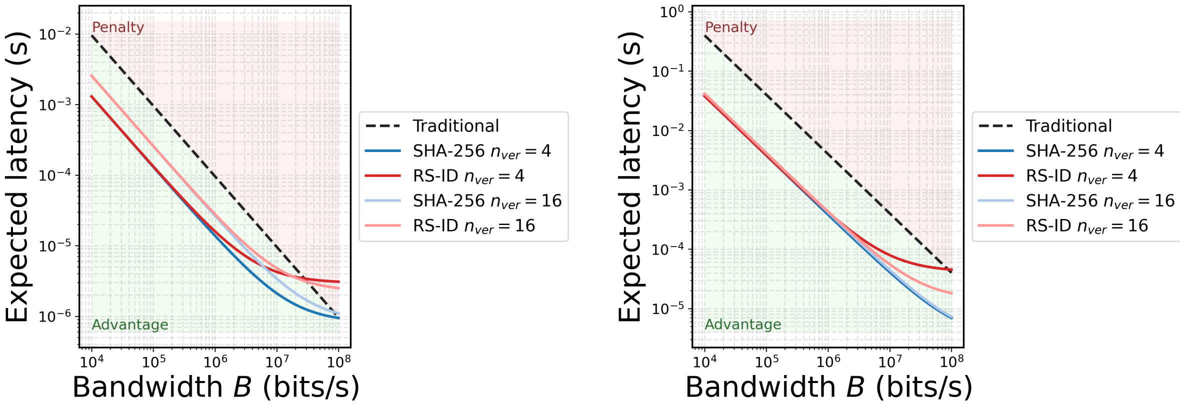
This theoretical bound yields a collision probability of approximately 6.25×10^{-2} for $n_{ver} = 4$ and 1.53×10^{-5} for $n_{ver} = 16$.

5.2 End-to-End Latency Evaluation

The measured parameters are integrated into the hybrid latency model to evaluate system performance under varying network constraints.

5.2.1 Bandwidth Sensitivity Analysis

Figure 5.2 analyzes the expected latency across a bandwidth spectrum from 10 kbps to 10 Mbps.



(a) Small payload (96 bits): cross-over at $\approx$ 2 Mbps.

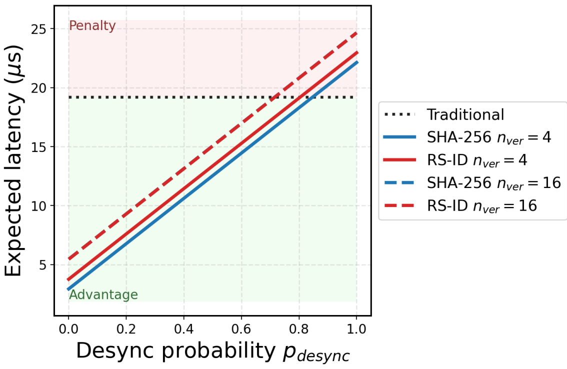
(b) Large payload (4001 bits): ID advantage persists $>$ 10 Mbps.

Figure 5.2: End-to-end latency vs. channel bandwidth ($p_{desync} = 0.1$). The plots reveal the break-even point where the computational cost of ID (t_{ver}) intersects with the transmission cost of the baseline. In the low-bandwidth case ($B < 1$ Mbps), ID offers up to 10$\times$ latency reduction.

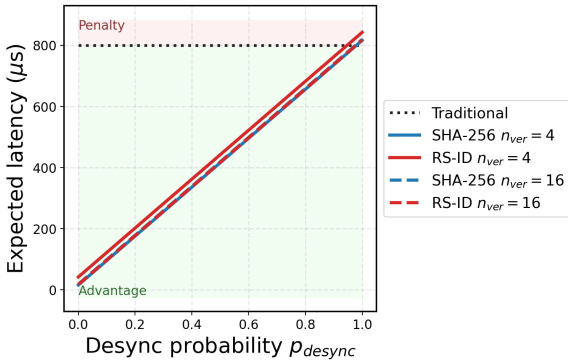
The results illustrate the fundamental trade-off between computation and communication. In the transmission-limited case ($B < 1$ Mbps), latency is dominated by the physical transmission time (T_{trans}), and ID outperforms the baseline by compressing the required transmission from n_{data} to n_{ver}. In the compute-limited case ($B > 10$ Mbps), transmission becomes negligible and the latency lower bound is set by the encoding time t_{ver}; since the traditional scheme has zero encoding overhead, it eventually becomes faster. For large payloads (Figure 5.2b), the break-even point shifts to higher bandwidths, indicating that ID-based synchronization becomes increasingly valuable as data complexity grows.

5.2.2 Perception Sensitivity Analysis

Figure 5.3 illustrates the system latency as a function of the state desynchronization probability (p_{desync}), measured at a fixed bandwidth of 5 Mbps. In this context, p_{desync} represents the frequency at which the robot's local sensor perception deviates from the authoritative state maintained by the DT. Since the proposed protocol only triggers a full-state fallback transmission upon a verification failure, its average latency is tied to this perceptual consistency.



(a) Small Payload (96 bits)



(b) Large Payload (4001 bits)

Figure 5.3: Latency sensitivity to prediction error ($B = 5$ Mbps). The intersection with the traditional baseline indicates the p_{desync} range in which ID-based synchronization yields lower expected latency. For 4001-bit payloads (b) the cross-over shifts closer to $p_{desync} = 1$ compared to the 96-bit payloads (a).

The intersection points between the ID-based curves and the traditional transmission baseline delineate the operational viability of the protocol. This performance boundary is primarily governed by the state desynchronization probability p_{desync}, which determines the frequency of fallback events. When the system operates in a high-consistency case (low p_{desync}), ID-based synchronization yields significant latency reductions by substituting large payload transmissions with compact verifiers. However, as p_{desync} approaches unity, the cumulative cost of frequent fallback transmissions and the fixed encoding overhead t_{ver} renders the traditional scheme more efficient.

The relative advantage of the encoding mechanisms is further influenced by their respective computational profiles. Across the evaluated configurations, the higher encoding cost associated with RS-ID results in consistently elevated latency curves compared to SHA-256 for identical (n_{ver}, n_{data}) pairs. This computational penalty interacts dynamically with the payload size; for small telemetry payloads (96 bits), the increased bit-budget for larger verifiers can outweigh the algebraic gains. Conversely, for larger sensor payloads (4001 bits), the reduction in RS-ID symbol count becomes the dominant factor, allowing the protocol to benefit from larger n_{ver} despite the increased tag length.

Overall, the results suggest that the latency advantage of ID-based synchronization depends on payload complexity: at low payloads it is more sensitive to encoding overhead, whereas at high payloads it is dominated by transmission savings and becomes less sensitive to moderate increases in p_{desync}.

5.2.3 Payload Scalability Analysis

To evaluate the scalability of the proposed mechanisms for high-dimensional sensor data, we characterized the end-to-end latency across a payload spectrum ranging from 10^2 to 4×10^3 bits. This analysis, visualized in Figure 5.4, assumes a fixed verifier length $n_{ver} = 16$, a bandwidth of 5 Mbps, and a state desynchronization probability $p_{desync} = 0.1$.

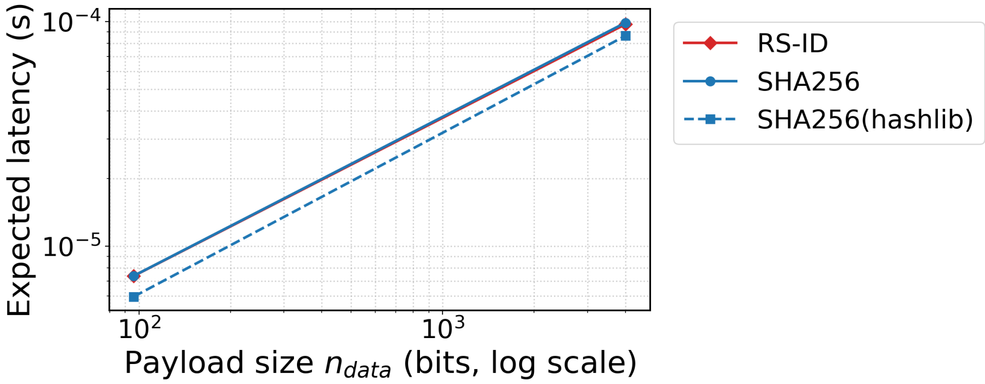


Figure 5.4: Expected end-to-end latency versus payload size. Evaluation at $n_{ver} = 16$ ($B = 5$ Mbps, $p_{desync} = 0.1$) comparing RS-ID and SHA-256. The linear scaling on the log-log plot illustrates the dominance of transmission time as n_{data} increases.

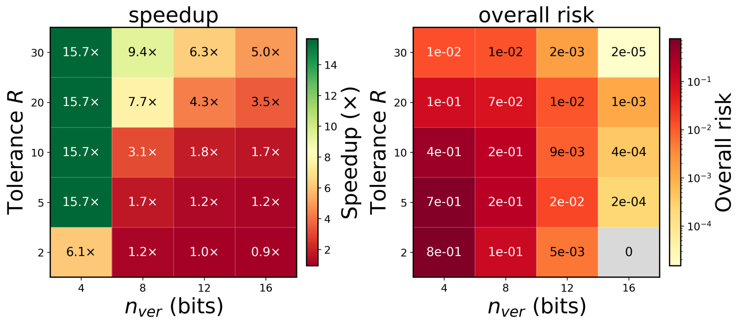
The experimental results show that for all tested mechanisms, the expected latency $E[T]$ scales linearly with the payload size n_{data} on a log-log scale. This linear relationship indicates that the transmission time, which is proportional to n_{data} during fallback events, remains the dominant component of the total latency as the data volume increases. Specifically, at $n_{ver} = 16$, the performance curves of RS-ID and the reference high-level backend of SHA-256 are nearly identical. Both mechanisms exhibit an expected latency of approximately 8.5×10^{-5} s for a 4001-bit payload.

A constant performance offset is observed for the optimized hashlib implementation of SHA-256. Across the entire evaluated spectrum, the optimized backend consistently maintains a lower latency profile compared to the other configurations.

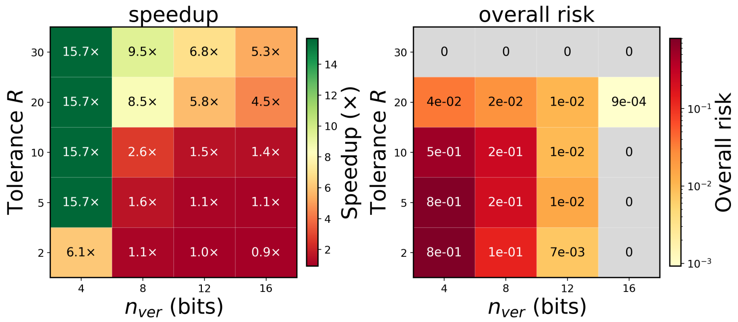
Overall, the results indicate that for larger payloads the end-to-end latency is mainly limited by fallback transmission, and differences between encoding mechanisms become secondary unless an optimized cryptographic backend is available.

5.3 K-ID: Quantifying the Safety-Performance Trade-off

Having established the baseline performance, we evaluate the K-ID extension, which introduces a semantic tolerance R to filter benign deviations. We sweep the tolerance threshold $R \in \{2, 5, 10, 20, 30\}$ and the verification strength $n_{ver} \in \{4, 8, 12, 16\}$ bits to map the interplay between physical and informational parameters.



(a) Gaussian model (SHA-256 truncation).



(b) Uniform model (SHA-256 truncation).

Figure 5.5: Heatmap summary of the K-ID grid over (R, n_{ver}). The left panel in each figure shows speedup (linear scale); the right panel shows empirical overall risk $P(\text{unsafe} \cap \text{accept})$ (log-scaled). Cells marked with "0" indicate an empirical zero (no failures observed in simulation).

The heatmaps in Figure 5.5 demonstrate that the tolerance radius R primarily determines the communication speedup, while the verifier length n_{ver} controls the safety floor. Analyzing the results vertically, increasing R consistently improves the speedup across all configurations. This trend is a direct result of semantic filtering, where a larger tolerance reduces the frequency of verification transmissions. Interestingly, the overall risk also tends to decrease as R increases. While a larger R expands the acceptance set K and thus increases the collision probability for any single check, this risk is outweighed by the rapid decline in the total number of transmission events. For the Uniform noise model, the risk drops to zero at $R = 30$ because the tolerance fully encapsulates the noise boundary ($a \approx 27.71$), making the protocol silent and eliminating collision opportunities. In the Gaussian model, the long-tailed distribution ensures that transmission triggers persist even at $R = 30$, leaving a residual risk that must be managed by the verifier bit-budget.

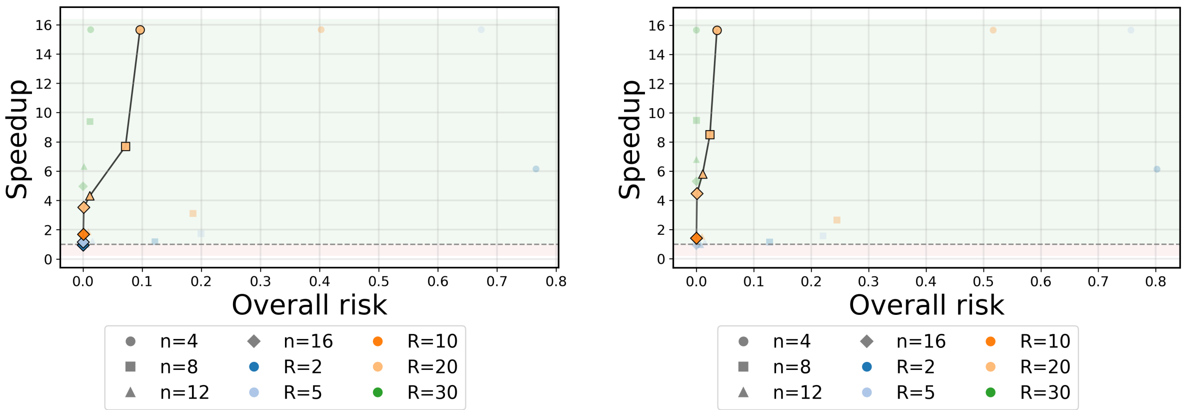
Horizontal analysis shows that increasing n_{ver} from 4 to 16 bits leads to a minor reduction in speedup due to the marginal bit-overhead per packet. However, the safety benefit is significant, with the overall risk exhibiting exponential decay as more bits are allocated to the verifier. For a tight tolerance of $R = 2$, increasing the verifier length suppresses the risk in the Gaussian model from 0.8 down to an empirical zero in our simulation runs. This confirms that increasing the verifier's bit-depth is the primary mechanism for lowering the safety floor, especially when narrow tolerances necessitate frequent active monitoring.

In conclusion, K-ID supports two distinct operating modes. When R is large, the protocol can stay silent within a tolerated error zone, and the efficiency gains depend on the noise distribution.

When n_{ver} is large, safety improves through probabilistic filtering, which remains effective even during frequent verification events. These findings suggest that systems in predictable environments can use larger R to reduce bandwidth, whereas systems facing uncertain or heavy-tailed disturbances require larger n_{ver} to maintain synchronization integrity.

5.4 Pareto Optimization for Deployment

To support deployment decisions, the multi-dimensional configuration grid is projected onto a speedup-versus-risk plane. Figure 5.6 visualizes the Pareto frontier for both Gaussian and Uniform error models, representing the set of non-dominated configurations where no further increase in speedup is possible without increasing the empirical risk. The specific values for these points are detailed in Table 5.2 and Table 5.3.



(a) Gaussian error model.

(b) Uniform error model.

Figure 5.6: Pareto frontiers for the K-ID protocol. The curves illustrate the trade-off between empirical risk and speedup. The absence of loose tolerance configurations (e.g., $R = 30$) from the non-dominated set reflects diminishing returns in speedup relative to the corresponding increase in acceptance-set size.

Table 5.2: Pareto-optimal configurations (Gaussian error model).

n_{ver}	R	Speedup	Overall risk
16	2	0.95	0.0
16	5	1.16	0.00023
16	10	1.69	0.0004
16	20	3.52	0.001
12	20	4.31	0.01
8	20	7.69	0.07
4	20	15.66	0.1

Table 5.3: Pareto-optimal configurations (Uniform error model).

n_{ver}	R	Speedup	Overall risk
16	10	1.42	0.0
16	20	4.47	0.00092
12	20	5.81	0.01
8	20	8.50	0.02
4	20	15.66	0.04

The Pareto frontiers reveal that the communication speedup tends to saturate as the tolerance radius R increases. While the experimental sweep included $R = 30$, these configurations are absent from the optimal frontier. At $R = 20$, the protocol effectively filters the majority of state deviations, and the total latency reaches a lower bound determined by the fixed computation time and verifier transmission. Further increasing the radius to $R = 30$ provides almost no additional bandwidth savings; instead, it increases the size of the acceptance set K, which elevates the risk of false positives. For the evaluated noise levels, $R = 20$ represents a clear knee point where the marginal benefit of increasing tolerance diminishes.

The shape of the frontier is also sensitive to the noise distribution. In the Gaussian case, the long-tail effect leads to a gradual increase in risk, with a residual risk of 0.001 remaining even at $R = 20$. In contrast, the Uniform model has a bounded support; once R reaches 10, the observed risk drops to zero within simulation limits, allowing for higher efficiency at a lower risk level.

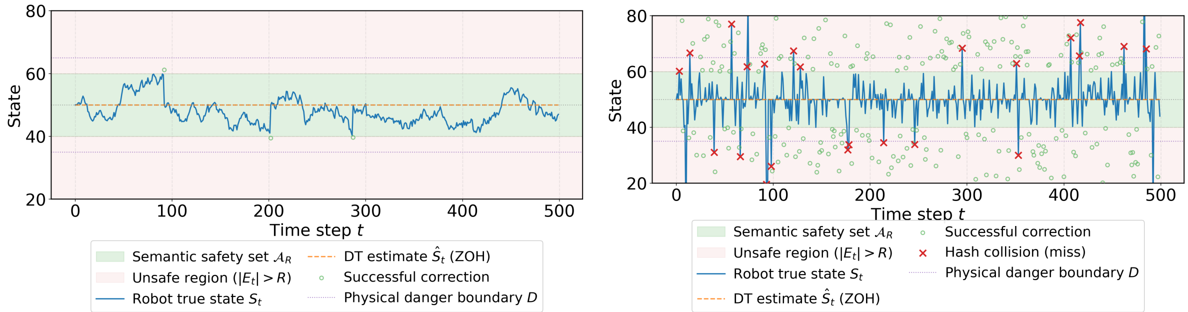
As a specific operating point, the configuration $(n_{ver} = 12, R = 20)$ achieves a $4\times$ to $6\times$ speedup with an empirical risk of approximately 1%. For systems requiring higher safety, $(n_{ver} = 16, R = 20)$ reduces the risk to the 10^{-3} level while still maintaining significant bandwidth savings.

5.5 Dynamic Evaluation Results: Random Walk with Drift

Building on the experimental design detailed in Section 4, this section presents the evaluation of the K-ID protocol under dynamic conditions. It first analyzes the qualitative behavior of the protocol through a representative stress-test trace, followed by a quantitative statistical analysis of safety risks and bandwidth efficiency across the parameter space.

5.5.1 Qualitative Analysis: The Anatomy of Failure

To visualize the system's behavior under dynamic drift, Figure 5.7 plots a 500-tick snapshot of the state evolution for two noise regimes ($R = 10$, $\sigma \in \{1, 16\}$). The blue curve represents the robot's true state S_t undergoing a random walk, while the dashed orange line indicates the DT's estimate $\hat{S}_t$. The green shaded band denotes the semantic safety set $\mathcal{A}_R$.



(a) $\sigma = 1.0$ (b) $\sigma = 16.0$

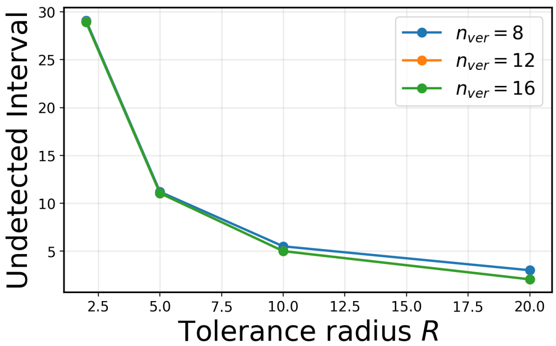
Figure 5.7: Drift-and-escape visualization for the dynamic random walk model ($R = 10$) over 500 ticks. The robot state S_t (blue) performs a random walk while the DT estimate $\hat{S}_t$ (orange) holds constant. Light-green circles mark successful corrections; red crosses mark collision-induced misses (modeled with $p_{\text{miss}} = 0.1$ per verification) where the boundary breach is masked, allowing the trajectory to temporarily diverge beyond the safety set.

The trace illustrates the protocol's bandwidth suppression mechanism. For extended periods, the robot drifts within the safety envelope without triggering full-state recovery transmissions, masking minor fluctuations. The light-green circles indicate nominal operation, where a boundary breach triggers an immediate fallback and correction. The red crosses highlight the failure mode induced by probabilistic verification: collision-induced misses (modeled here with a per-verification miss probability of $p_{\text{miss}} = 0.1$) can mask boundary breaches. In these instances, the protocol does not detect the excursion, allowing the robot to "escape" the safety set for a short period.

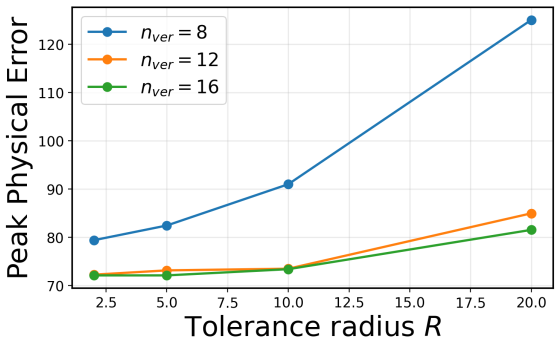
Comparing the two panels suggests that under mild drift ($\sigma = 1$), boundary interactions are sparse and escapes are rare within 500 ticks, so the protocol is mostly silent. Under heavy drift ($\sigma = 16$), boundary interactions become frequent and clustered, meaning that even a modest miss probability can produce longer periods of unobserved instability. This observation suggests that in dynamic settings, collision risk should be evaluated jointly with the drift scale and the rate of boundary encounters, rather than in isolation.

5.5.2 Quantitative Risk: Persistence and Magnitude

Beyond the single trace, we quantify the statistical risk profile using the parameter sweep data ($R \in \{2.5, 5, 10, 20\}$). Figure 5.8 reports the 95th-percentile summaries for Breach Duration (L_R) and Maximum Drift under a Gaussian random walk model.



(a) 95th percentile Breach Duration L_R.



(b) 95th percentile of Maximum Drift.

Figure 5.8: Dynamic risk metrics versus tolerance radius R (Gaussian drift $\sigma = 16.0$). Panel (a) shows the duration for which the desynchronization remains undetected once it exceeds R. Panel (b) captures the peak physical error during these undetected intervals.

Figure 5.8(a) illustrates a downward trend in the undetected interval L_R as the tolerance radius R increases. This behavior is tied to the stability of the drifting state relative to the threshold. At tight tolerances (e.g., $R = 2.5$), the system is sensitive to minor fluctuations, and the high correlation between consecutive steps in a random walk means that once the state crosses the boundary, it is likely to remain in a "breached" state for a sustained period. As R increases, the criterion for a breach becomes more stringent. While a larger R increases the induced acceptance-set size $K = |\mathcal{A}_R|$ (and thus the per-check miss probability $p_{miss} \approx K/2^{n_{ver}}$), extreme deviations are also statistically more likely to fluctuate back within the tolerance due to the stochastic nature of the drift, thereby terminating the recorded breach duration earlier.

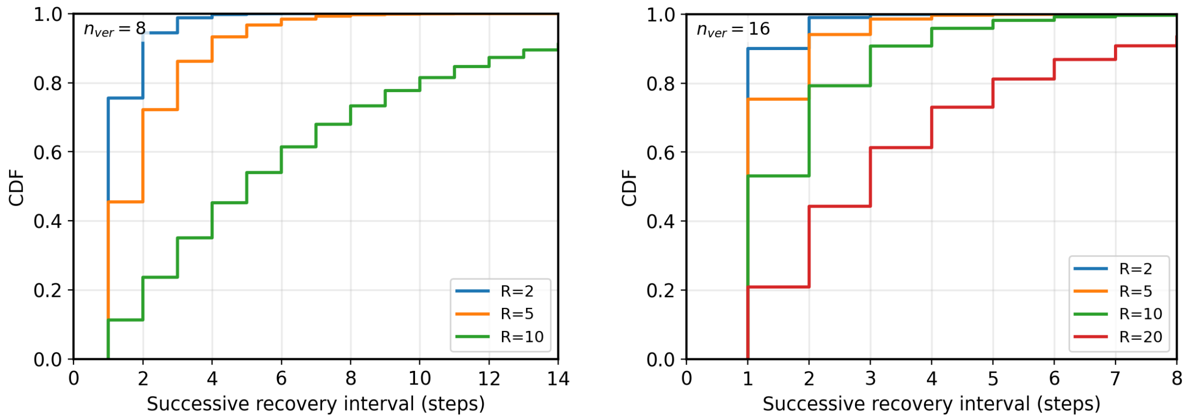
The curves for $n_{ver} \in \{8, 12, 16\}$ appear close because, in this setting, the per-event miss probabilities are sufficiently low that the duration is primarily governed by the underlying random walk dynamics rather than by collision-induced cascades. The exclusion of $n_{ver} = 4$ is necessary because its miss probability is close to one for moderate R, which makes the metric dominated by verifier misses and not informative for comparing practical configurations.

Figure 5.8(b) illustrates the physical magnitude of these failures. Despite the decrease in duration shown in Figure 5.8(a), the peak physical error scales consistently with R. This reflects the masking effect: a larger tolerance allows a greater magnitude of drift to accumulate before any verification is triggered. Furthermore, the $n_{ver} = 8$ configuration exhibits a higher peak error compared to $n_{ver} \geq 12$, indicating that weaker verifiers are more susceptible to consecutive misses once the system enters the $|E_t| > R$ condition. This extends the time window for the drift to grow unchecked. The convergence of the $n_{ver} = 12$ and $n_{ver} = 16$ curves suggests that for this noise level, a 12-bit verifier already reaches a point of diminishing returns where the peak error is dominated by the masking radius R rather than the probability of hash collisions.

Overall, the results show a trade-off between the duration and the magnitude of synchronization failures. Increasing R reduces the average time spent in an undetected state, but it leads to larger physical errors when failures occur.

5.5.3 Throughput Analysis: Full-State Update Intervals

This section evaluates the protocol's effective throughput by analyzing the SRI, defined as the number of time steps between two consecutive successful state synchronizations. As illustrated in the data flow (Figure 3.3), a state restoration is not merely triggered by the physical drift exceeding the threshold ($|E_t| > R$), but is also contingent upon the verifier successfully identifying the desynchronization to trigger the n_{data} fallback. Consequently, the SRI distribution directly quantifies the actual re-synchronization rate under combined physical and cryptographic constraints.



(a) $n_{ver} = 8$.

(b) $n_{ver} = 16$.

Figure 5.9: Empirical Cumulative Distribution Function (CDF) of SRIs for the random-walk drift model (Gaussian drift $\sigma = 16.0$, quantization $\Delta = 0.1$). The x-axis is truncated at 15 ticks to highlight the primary synchronization intervals. The curves shift to the right as the tolerance radius R increases, while the verifier length n_{ver} governs the distribution slope and the suppression of long-term drift accumulation.

Figure 5.9 reports the empirical CDF of the SRI across varying tolerances. For the $n_{ver} = 8$ configuration, the $R = 20$ case is excluded because the verifier space is insufficient to accommodate the required state tolerance. At a resolution of $\Delta = 0.1$, the acceptance set cardinality $K = |\mathcal{A}_{R=20}| = 401$ exceeds the 256 available states of an 8-bit verifier. In this setting, the verifier cannot reliably distinguish between synchronized and drifted states, so recovery is not triggered reliably.

A comparison between Figure 5.9(a) and (b) reveals how different parameter settings influence synchronization behavior. Within each sub-figure, increasing R shifts the CDF curves to the right, reflecting a reduced update frequency. A larger physical tolerance reduces the frequency of boundary crossings, thereby extending the typical interval between full-state updates. However, the 8-bit verifier exhibits a significantly shallower slope and a heavier tail compared to the 16-bit version. This gap indicates that even when the drift exceeds R, frequent verifier collisions (false positives) occur, which delays state recovery and forces the system to wait for subsequent time steps to resolve the error.

In contrast, the 16-bit verifier curves converge rapidly toward 1.0 after crossing their respective R thresholds. By reducing the probability of collisions to negligible levels, the SRI distribution becomes almost entirely governed by the stochastic timing of the physical drift rather than by verification failures. This demonstrates that increasing the verifier bit-budget effectively eliminates the uncertainty in recovery timing.

Overall, the SRI distribution separates the roles of the parameters in the K-ID protocol: the tolerance radius R primarily sets the typical spacing between recoveries, while the verifier length n_{ver} determines how reliably a breach is detected once the error bound is exceeded. Since the acceptance set K grows with R and finer quantization Δ, the verifier bit-width must be scaled accordingly to avoid extended synchronization gaps caused by collision-induced recovery failures.

6 Conclusion and Future Work

6.1 Conclusion

This thesis investigates ID-based synchronization for edge-based DTs in bandwidth-constrained environments. Beyond reducing traffic volume, the ID mechanism can reduce end-to-end latency by replacing full payload transfer with "state verification". The results support the following conclusions:

First, the ID protocol reduces the sensitivity of synchronization latency to state dimension in bandwidth-limited settings. Experiments show that for $B < 1$ Mbps, ID yields lower end-to-end latency than full-state transmission. For high-dimensional sensor data such as LiDAR, the additional encoding overhead is typically outweighed by reduced fallback transmission cost, improving freshness by avoiding unnecessary payload transfers.

Second, for the evaluated software stack, truncated hashing based on SHA-256 is typically a more practical engineering choice than algebraic coding via RS-ID for edge deployments. The main driver is implementation efficiency: optimized SHA-256 backends achieve low and largely payload-independent encoding latency, whereas RS-ID incurs higher computation that depends on the symbolization of the payload.

RS-ID nevertheless remains relevant in two cases. The first is high-integrity applications where rigorous mathematical error bounds are mandatory (i.e., a provable upper bound on failure probability under the stated modeling assumptions). The second is transmission-congested, fallback-heavy operation, where full-state recovery is triggered frequently and the resulting transmission delay dominates the end-to-end latency; in that case, marginal compute-time differences between encoding mechanisms become comparatively less influential.

Third, K-ID implements the shift from "bitwise exactness" to "semantic compliance". The Pareto analysis shows that introducing a semantic tolerance radius R allows the system to trade reduced verification frequency (and thus reduced fallback traffic) against increased acceptance-set size. In the evaluated sweeps, a moderate tolerance improves speedup substantially, while configurations with very loose tolerance (e.g., $R = 30$) do not appear among the non-dominated points due to diminishing returns in speedup relative to the growth of the acceptance set.

As a practical guideline for the studied noise levels, $R \approx 20$ behaves as a knee point: further increases yield limited marginal speedup while reducing verifier selectivity. Here, selectivity refers to the verifier's physical sensitivity in distinguishing negligible drift (treated as semantically admissible) from critical deviations that should trigger recovery; as the acceptance set expands, a larger fraction of off-nominal states becomes indistinguishable from admissible noise at the verifier interface.

Fourth, the dynamic evaluation highlights time-correlated risk in the presence of probabilistic verification. Under random-walk drift, verifiers with insufficient bit-depth can transform isolated collisions into persistent, undetected boundary breaches once $|E_t| > R$. The analysis based on the SRI separates the roles of the parameters: the tolerance radius R mainly affects how often recovery is triggered, while the verifier bit-width n_{ver} affects how often a triggered recovery is missed due to collisions.

This also exposes a capacity-based feasibility constraint. Let $K \triangleq |\mathcal{A}_R|$ denote the acceptance-set size induced by the physical radius R at quantization step Δ (e.g., in 1D, $K = 2\lfloor R/\Delta \rfloor + 1$). Under the accept-set approximation, the miss probability scales as $p_{miss} \approx K/2^{n_{ver}}$. If K is comparable to or exceeds the available verifier space $2^{n_{ver}}$ (e.g., large R at fine Δ with short n_{ver}), the verifier space is effectively exhausted and reliable detection cannot be guaranteed. Finally, the dynamic metrics reveal an inherent trade-off: while increasing R can reduce the frequency of updates, it increases the maximum physical drift accumulated during undetected intervals.

Overall, the evaluation quantifies the coupled trade-offs among computation, communication, and control fidelity. The results yield design guidance for semantic synchronization policies, where the tolerance radius R, acceptance-set size K, verifier bit-width n_{ver}, and quantization Δ are selected jointly to satisfy both bandwidth and safety constraints.

6.2 Future Work

While this work has characterized the latency and reliability of K-ID, several avenues remain to extend its theoretical depth and practical applicability in next-generation networks.

6.2.1 Energy-Aware Identification Policies

Current evaluations utilize traffic volume and latency as proxies for system efficiency. However, for battery-powered Internet of Things (IoT) devices, energy consumption is often the dominant constraint. Future work should explicitly model the energy–delay product, balancing the computational energy cost of hashing against the transmission energy cost of radio interfaces. By integrating the First-Order Radio Model [7] with real-time power profiling of edge hardware, we aim to derive an energy-optimal policy that dynamically adapts the tolerance radius R (and thus the induced acceptance-set size K) and verifier length n_{ver} based on the device's remaining battery life.

6.2.2 Integration with 6G Physical Layer Security

The current implementation relies on synchronized PRNGs, which introduces a dependency on upper-layer clock synchronization protocols like PTP. To align with the architecture of 6G networks, future iterations will investigate the integration of PLKG. By exploiting the channel reciprocity in Time-Division Duplex (TDD) systems, the robot and the DT can extract common randomness directly from the Channel State Information (CSI). This approach would turn channel noise into a synchronization resource, eliminating the need for clock synchronization and enhancing robustness against timing attacks in harsh industrial environments.

6.2.3 Scalability to Multi-Agent Systems

This thesis focused on a single robot–DT loop. However, industrial scenarios often involve swarms of robots sharing a physical workspace. A critical extension is the development of semantic multiple access schemes, where multiple robots superimpose their identification codes on a shared channel. Future work will explore algebraic constructions that allow the DT to verify the aggregate safety of the swarm directly in the coded domain, checking for "Semantic Inter-Robot Collisions" without decoding individual states, thereby scaling the K-ID approach to massive multi-agent systems.

6.2.4 Predictive DTs and Adaptive K-ID

The dynamic drift evaluation in this thesis uses a deliberately simple predictor (ZOH) to isolate protocol-induced effects. A natural next step is to integrate stronger predictors such as Kalman filtering or learned kinematic models, and to study how prediction quality reshapes both the masking risk (through the breach rate) and the collision risk (through the verification rate). In addition, future work should explore adaptive policies that tune (R, n_{ver}) online based on observed breach statistics and safety budgets, tightening the tolerance under persistent drift and relaxing it during stable operation to reduce bandwidth.

6.2.5 Decoder Compute, Scheduling, and Quantization Sensitivity

Beyond bandwidth and latency, practical deployments must also account for DT-side compute and memory. Since the acceptance-set size scales as $K = |\mathcal{A}_R| \approx 2\lfloor R/\Delta \rfloor + 1$ in 1D, a direct implementation precomputes $\mathcal{V}_R$ by hashing $\mathcal{O}(K)$ states per DT update, and stores up to $|\mathcal{V}_R| \leq K$ verifier codewords. Future work should therefore characterize end-to-end performance under explicit decoder compute constraints (e.g., bounded hash evaluations per tick), and study algorithmic accelerations (incremental updates, caching, Graphics Processing Unit (GPU) offload).

In addition, the dynamic setting naturally suggests a scheduling-centric view: rather than only reporting per-tick risk, one can characterize the distribution of full-state update intervals (e.g., via SRIs, survival functions, and CDF metrics) to answer questions like "with 99% probability, how long does the channel remain idle?" Finally, because Δ is tied to sensor precision and DT compute budget, a systematic sensitivity analysis over Δ is needed to guide robust configuration across heterogeneous sensing modalities.

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